

★ WEEK 5 · PART 1 of 3

ARCH: Modeling Conditional Variance

You Can't Forecast Direction — But You Can Forecast Risk

Conditional vs unconditional variance · ARCH(q) · the clustering engine

Weeks 3–4 modeled the mean and found returns nearly unpredictable. The structure lives in the variance. Part 1 introduces the idea that finally captures it: a variance that changes over time.

1

PART 1 • ARCH

When Variance Isn't Constant

The conditional-variance idea, the ARCH model that first captured volatility clustering, and the fat tails and diagnostics that come with it.

The Economist Who Modeled Fear — Part 1

How a question about inflation changed risk forever

Robert Engle asked a simple question: is the UNCERTAINTY of inflation itself changing over time? Standard models assumed a fixed variance — and he suspected they were wrong.

The setting

- Late 1970s inflation was volatile, then calm, then volatile again.
- Every model of the day assumed a single, constant variance.
- But the uncertainty itself seemed to cluster in time.

The insight

- What if variance depends on the recent past, like the mean can?
- Then risk is forecastable even when direction is not.
- That question became the ARCH model.

The Economist Who Modeled Fear — Part 2

ARCH is born, and the lesson

Engle's 1982 ARCH model let the variance evolve with recent shocks. It reshaped risk management and earned him the 2003 Nobel — shared with Granger, of cointegration fame.

What it changed

- Volatility became something you could estimate and forecast.
- Value-at-Risk, option pricing, and risk desks were rebuilt on it.
- It is now the default lens for financial risk.

The lesson for us

- Give up on forecasting direction; forecast risk instead.
- Model the variance as its own dynamic process.
- This is where the predictable structure in returns lives.

From the Mean to the Variance

Three ideas that set up this week

1

The mean of returns is nearly unpredictable

Weeks 3–4 showed ARIMA finds almost no linear structure in returns. Their ACF is essentially zero — direction bets have no statistical edge.

2

But their VARIANCE is highly predictable

Squared returns are strongly autocorrelated: big moves cluster. That is the stylized fact of Week 1, and the thing ARCH was built to model.

3

So we switch targets: model the variance

Keep the mean simple, and put the dynamics in the variance. That is the entire ARCH/GARCH program.

Week 5 in one line: you cannot forecast the return, but you can forecast its risk.

Why Constant Variance Fails

Markets are not equally risky every day

Every model so far assumed one fixed variance for all time. But real markets alternate between calm and crisis — a single number cannot describe both a quiet Tuesday and a crash.

What we observe

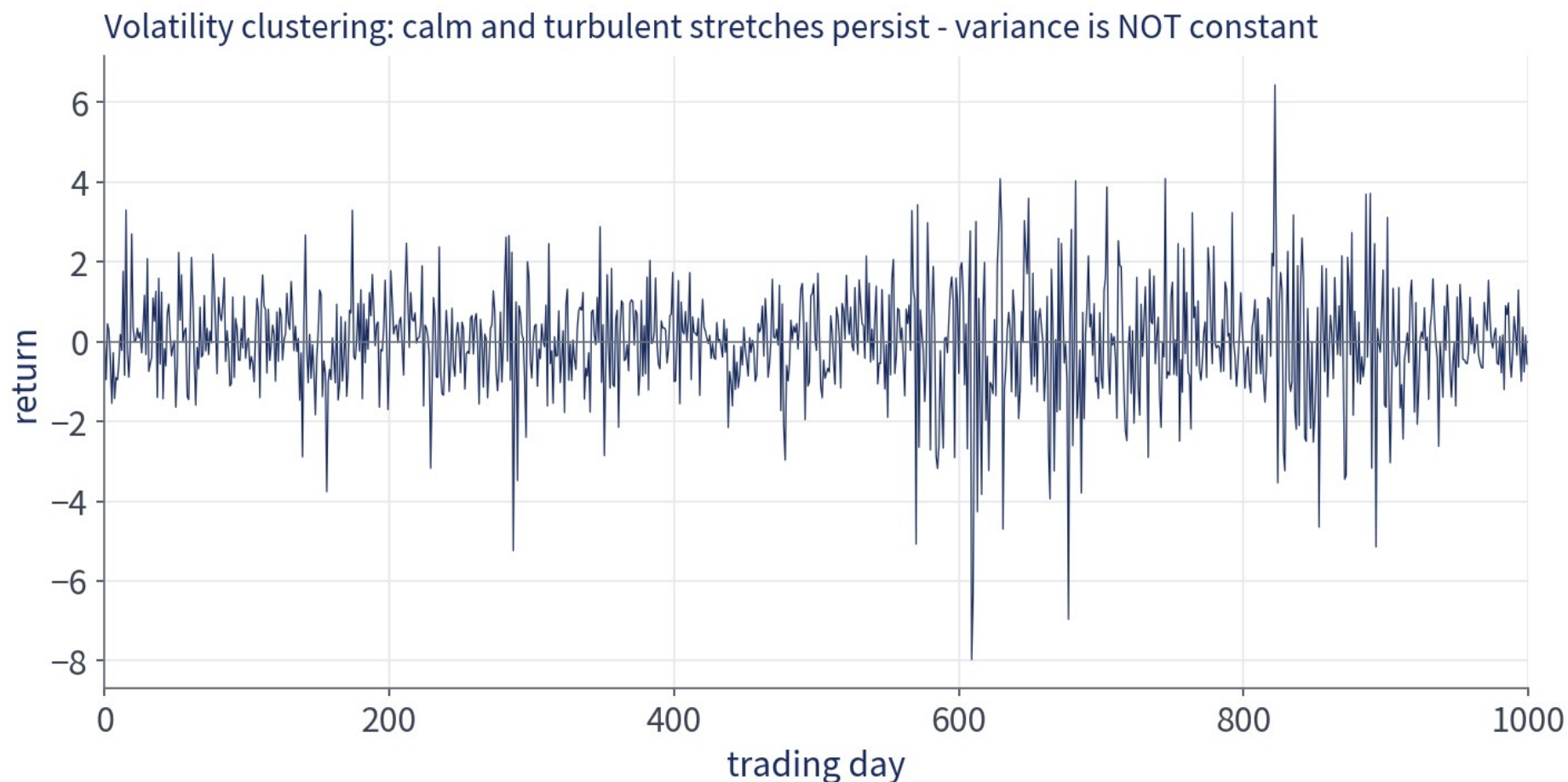
- Quiet weeks of tiny moves, then bursts of violent ones.
- The turbulence persists — clustering, not random spikes.
- A constant-variance model is right on average, wrong every day.

Why it matters

- Risk limits set on average variance blow up in a crisis.
- Option prices and margins need TODAY's risk, not the average.
- We need a variance that adapts — day by day.

Volatility Clustering, One More Time

Calm and turbulent stretches each persist — variance is plainly not constant. This is the pattern (Week 1) that a constant-variance model cannot represent.



Conditional vs Unconditional Variance

The single most important idea this week

$$\underbrace{\text{Var}(r_t) = \sigma^2}_{\text{unconditional: fixed}}$$

$$\underbrace{\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2}_{\text{conditional: time-varying}}$$

Unconditional variance

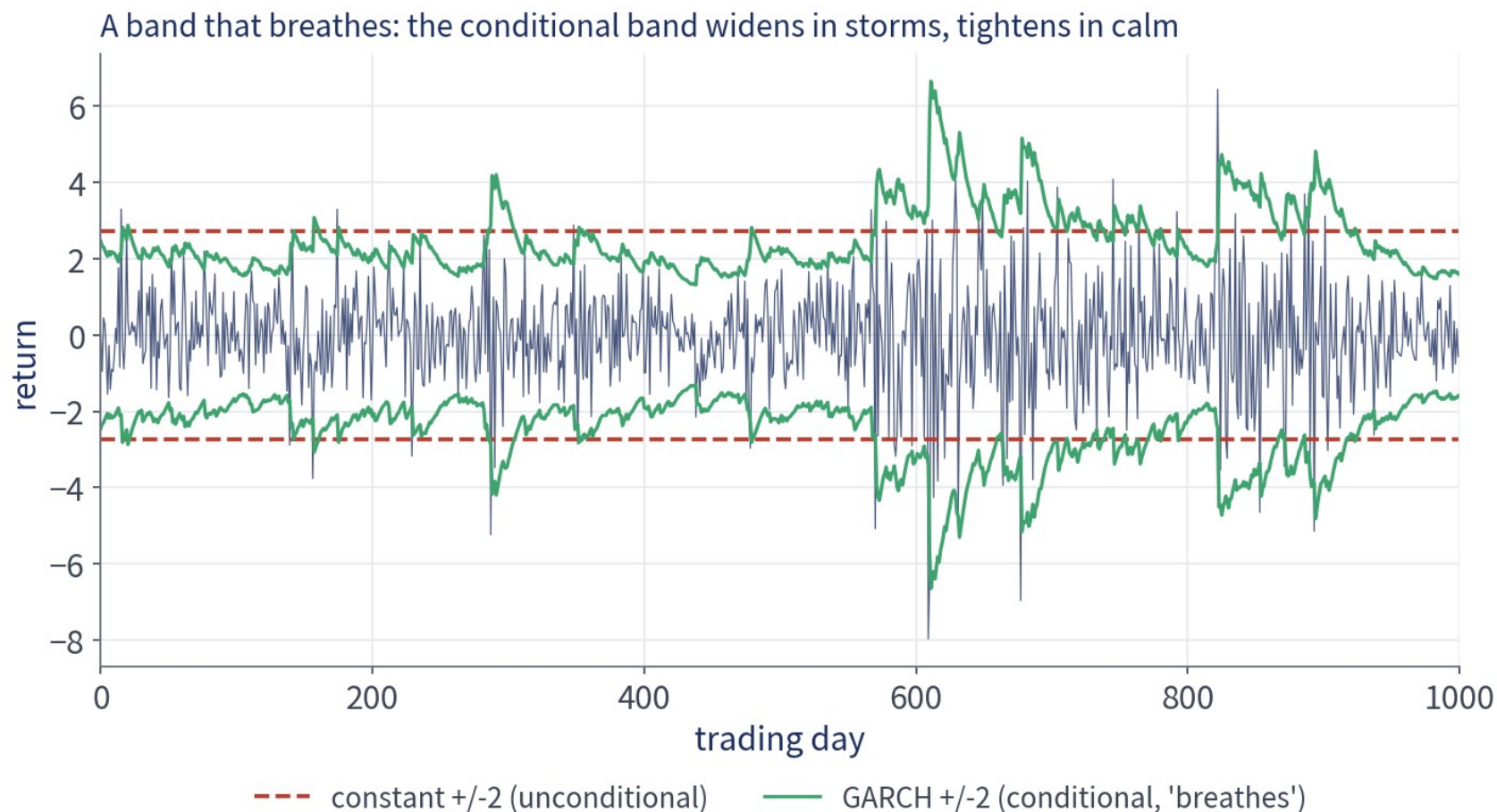
- The long-run, overall variance — one fixed number.
- It ignores what just happened in the market.
- Useful as an average, useless for today's risk.

Conditional variance

- The variance GIVEN the recent past — it changes each day.
- It is large after big shocks, small after calm.
- This time-varying number is what ARCH models.

A Band That Breathes With the Market

The constant (unconditional) band is the same every day. The conditional band widens in storms and tightens in calm — this is the risk number you actually want.



Two Equations: Mean and Variance

ARCH models the second one

A volatility model has two parts: a MEAN equation (often just a constant, since returns are unpredictable) and a VARIANCE equation that lets today's risk depend on the past.

The mean equation

- Usually as simple as 'return = constant + shock'.
- We already know there is little to model here (Weeks 3–4).
- Sometimes a small ARMA, but often nothing at all.

The variance equation

- This is where ARCH/GARCH does its work.
- It makes the shock's SIZE depend on recent shocks.
- All the modeling effort now goes here.

Separating Size From Direction

Every return is a size times a direction

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{iid}{\sim} (0, 1)$$

Reading it

- The shock splits into a time-varying size and a random sign.
- σ_t is the conditional standard deviation — the 'size'.
- z_t is a standardized draw with mean 0 and variance 1 — the 'direction'.

Why this split

- It isolates exactly the piece ARCH models: σ_t .
- The direction z_t stays unpredictable (efficient markets).
- Model the size; leave the sign to chance.

The ARCH(1) Model

Variance driven by the last squared shock

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 \quad (\omega > 0, \alpha \geq 0) \quad \text{ARCH(1)}$$

How to read it

- Today's variance is a baseline ω plus a reaction to the last shock.
- A big shock yesterday (large squared value) $\rightarrow$ big variance today.
- The parameter α sets how strongly shocks feed into variance.

Why squared

- Squaring makes both up and down shocks raise variance.
- Only the SIZE of the shock matters, not its sign.
- That is what produces clustering of large moves.

The Engine of Clustering

How one shock echoes into tomorrow's risk

A large shock today enlarges tomorrow's variance, which makes another large shock more likely, which enlarges the day-after variance — a self-reinforcing loop that produces clusters.

The feedback

- Big shock $\rightarrow$ higher next variance $\rightarrow$ likely another big shock.
- Calm $\rightarrow$ lower next variance $\rightarrow$ likely more calm.
- Turbulence and tranquility both become sticky.

The balance

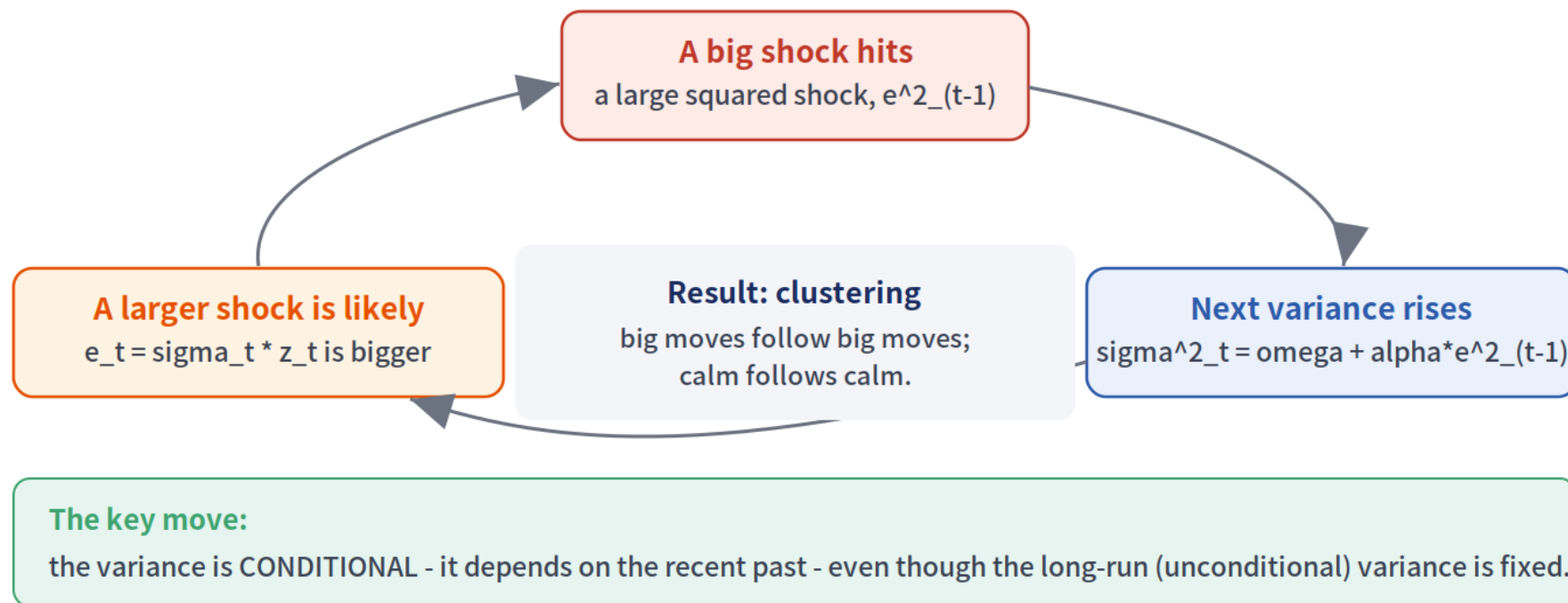
- ω anchors the long-run level of variance.
- α controls how twitchy the variance is to shocks.
- Too large an α and the variance explodes — bounds apply.

The Clustering Feedback Loop

ARCH ties today's variance to yesterday's shock. That single link turns isolated shocks into persistent clusters — while the long-run variance stays fixed.

The Engine of Volatility Clustering

ARCH makes today's variance depend on yesterday's shock - a self-reinforcing loop.



The ARCH(q) Model

Let the last q shocks all have a say

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 \quad \text{ARCH}(q)$$

The extension

- Variance depends on the last q squared shocks, each weighted.
- q is the memory: how many days of shocks still matter.
- ARCH(1) is the special case $q = 1$.

The practical snag

- Real volatility has long memory — clusters last weeks.
- Capturing that with ARCH needs a large q (many parameters).
- That inefficiency is exactly what GARCH will fix (Part 2).

Unconditional Variance and Stationarity

When is an ARCH model well-behaved?

$$\mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \sum_i \alpha_i} \quad (\text{finite iff } \sum_i \alpha_i < 1)$$

The long-run variance

- Averaged over all time, the variance settles to $\omega/(1 - \sum \alpha)$.
- This is the unconditional variance the model reverts toward.
- The conditional variance orbits around this level.

The stationarity condition

- The α 's must sum to less than 1 for a finite variance.
- If they sum to 1 or more, variance grows without bound.
- So the reaction coefficients are always constrained.

Fat Tails, for Free

ARCH generates leptokurtosis by itself

Even with normal shocks z_t , mixing calm and turbulent periods produces returns with fatter tails than a normal — ARCH reproduces a stylized fact of Week 1 automatically.

Why it happens

- Some days draw from a wide variance, others from a narrow one.
- This mixture of variances has heavier tails than any single one.
- So extreme returns arise more often than a normal predicts.

The catch

- The tails ARCH makes are still not fat ENOUGH for real data.
- Real returns are heavier-tailed than plain ARCH implies.
- The fix is heavier-tailed shocks (Student-t) — Part 2.

ARCH and the Stylized Facts

One model, several facts at once

Facts it reproduces

- Volatility clustering — its whole reason for being.
- Fat tails — from the mixture of variances.
- Uncorrelated returns with correlated squared returns.

Why that is compelling

- A single mechanism explains several separate observations.
- That is the mark of a genuinely good model.
- It ties Week 1's facts directly to a workable equation.

Testing for ARCH Effects

Is there volatility to model at all?

ARCH-LM test: regress ε_t^2 on $\varepsilon_{t-1}^2, \dots, \varepsilon_{t-q}^2$; H_0 : no ARCH

How to check

- Look at the ACF of SQUARED returns — spikes mean ARCH effects.
- Formally, run the ARCH-LM test on the squared residuals.
- A small p-value says: yes, the variance has structure.

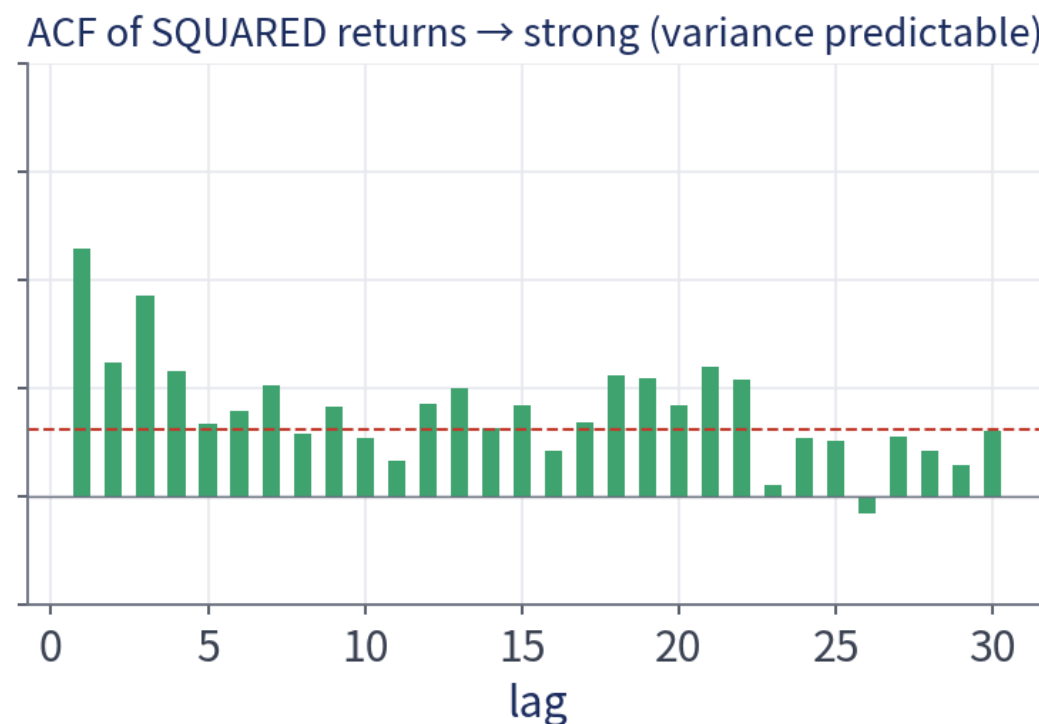
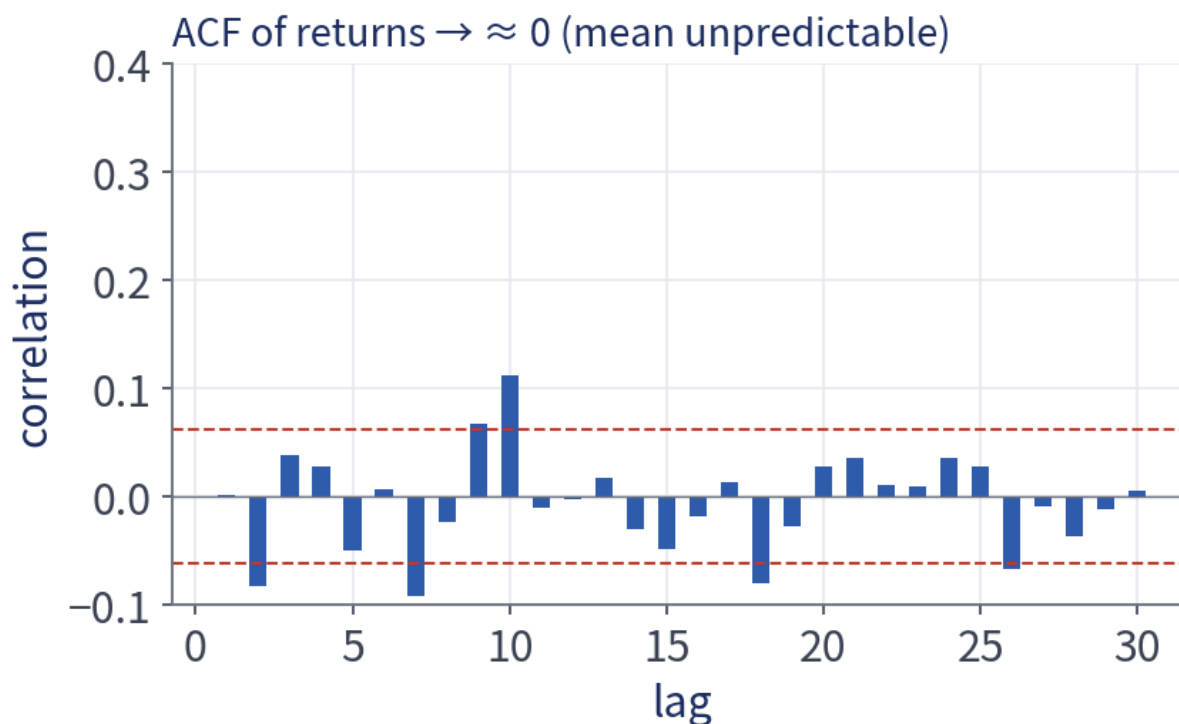
When to apply it

- Always test before fitting — confirm there is clustering.
- Also test the RESIDUALS after fitting your mean model.
- Leftover ARCH effects mean you need a volatility model.

The ARCH Signature

Returns are uncorrelated (left), yet squared returns are strongly autocorrelated (right). That gap is the fingerprint of ARCH effects — predictable variance.

The ARCH signature: squared returns are autocorrelated even when returns are not



Why ARCH Alone Isn't Enough

A great idea that needs one more step

ARCH captures clustering, but to match volatility's long memory it needs many lags and many parameters. GARCH will fix this with one elegant addition — memory of past variance.

The problem

- Real volatility clusters persist for weeks.
- Pure ARCH needs a high q to reach that far back.
- Many parameters mean noisy, over-fit estimates.

The preview

- GARCH adds a term for YESTERDAY's variance.
- One extra parameter captures long memory compactly.
- GARCH(1,1) does what ARCH(10+) struggles to — next part.

Building an ARCH Model

From returns to a fitted variance

1 • Mean

fit & remove

Fit a simple mean model;
keep the residuals.

2 • Test

ARCH-LM

Confirm the squared residuals show clustering.

3 • Specify

pick q

Choose the ARCH order from the squared-return ACF.

4 • Estimate

fit variance

Fit the ARCH by maximum likelihood.

5 • Check

standardize

Standardized residuals should be clean, no ARCH left.

The same discipline as Weeks 3–4, now applied to the variance instead of the mean.

Detect and Fit ARCH

Direct Claude Code from clustering to a variance model

GOAL Confirm ARCH effects in real returns, fit an ARCH model, and visualize the fitted conditional variance.

[INPUT] Prompt · Claude Code

Write and run `arch_lab.py` that:

1. downloads daily returns for a ticker and plots the ACF of returns vs squared returns;
2. runs the ARCH-LM test and reports whether ARCH effects are present;
3. fits an ARCH(q) with the 'arch' library (try $q = 1$ and a few larger q);
4. overlays the fitted conditional ± 2 -sigma band on the returns.

Tell me if the standardized residuals still show any ARCH structure.

How to work with it

- Expect flat return ACF, strong squared-return ACF.
- The band should widen in turbulent stretches.
- Larger q helps — but watch the parameter count.
- Keep it; Part 2 upgrades ARCH to GARCH.

Deliverable: `arch_lab.py` + a conditional-variance band over your returns.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

Returns are uncorrelated but squared returns are not. What does this tell you?

ARCH effects present

Q2

What is the difference between conditional and unconditional variance?

Time-varying vs fixed

Q3

Why does pure ARCH need so many lags for real data?

Long memory in vol

All three come from the conditional-variance idea and the clustering engine.

Three Misconceptions to Avoid

Where volatility modeling trips people up

Common myths

- "ARCH lets me forecast the direction of returns."
- "Volatility clustering means returns are autocorrelated."
- "A constant variance is fine most of the time."

The reality

- ARCH forecasts the SIZE of moves, never their sign.
- Returns stay uncorrelated; it is their SQUARES that cluster.
- Constant variance is wrong exactly when it matters — in a crisis.

Three Things to Carry Forward

Part 1 in one screen

1

Variance is the target now, not the mean

Returns are unpredictable in direction but predictable in size. ARCH models the conditional variance — today's risk, given the recent past.

2

ARCH ties variance to recent squared shocks

A big shock raises tomorrow's variance, creating the self-reinforcing loop that produces clustering — and fat tails as a bonus.

3

ARCH works, but is inefficient

Matching volatility's long memory needs many ARCH lags. One extra idea — memory of past variance — fixes that. Enter GARCH.

Next — Part 2: GARCH, the workhorse that adds variance memory and rules the world's risk desks.

★ WEEK 5 · PART 1 · PRACTICE

The ARCH Model, Worked by Hand

Turn One Shock Into Tomorrow's Risk, Then Test For It

The ARCH(1) recursion, stationarity, and the LM test

Part 1's core idea, with the arithmetic shown. We run the ARCH(1) recursion so a big shock spikes the next day's variance and then decays - volatility clustering - then check stationarity through the unconditional variance and test whether ARCH effects are there to model at all.

1P

PART 1 · APPLICATIONS

When Yesterday's Shock Sets Today's Risk

Case A: run the ARCH(1) recursion and watch a big shock spike tomorrow's variance, then fade - the clustering that constant-variance models miss. Case B: check stationarity through the unconditional variance and test for ARCH effects with the LM test. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Variance is conditional, not constant

Markets aren't equally risky every day. ARCH lets today's variance depend on yesterday's shock, so risk clusters.

2

One squared shock drives the next variance

The ARCH(1) recursion adds a fraction of yesterday's squared shock to a baseline - a big move today means a risky tomorrow.

3

Check stationarity, and test first

The unconditional variance exists only if the coefficient is below one, and an LM test confirms there are ARCH effects to model.

Everything here is a calculator exercise: one recursion, one unconditional variance, one test.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the ARCH parameters and shocks.
- Run the recursion, or test and check stability.
- Then read the risk.

What you should watch

- How a big shock spikes the next variance.
- How the spike decays back toward the mean.
- How the LM test flags clustering.

The ARCH(1) Recursion

One shock, turbulent market
A calm market can turn turbulent overnight after a big move.

ARCH(1) captures that: today's variance is a baseline plus a fraction of yesterday's squared shock. We run the recursion through a sequence that includes a crash-sized shock, and watch the conditional variance spike, then decay back toward its long-run

The model

- Baseline omega of 0.5.
- Coefficient alpha of 0.6 on the last shock.
- Return equals volatility times a normal draw.

What we'll do by hand

- Apply the recursion to a calm shock.
- Apply it to a big shock.
- Watch the spike decay.

The ARCH(1) Model

Variance from yesterday's squared shock

$$r_t = \sigma_t z_t, \quad z_t \sim N(0, 1);$$

$$\underbrace{\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2}_{\text{today's variance from yesterday's shock}}$$

today's variance from yesterday's shock

Size from direction

- Every return is volatility times a direction.
- Squaring the shock strips the sign.
- Only the magnitude drives risk.

The clustering link

- A big squared shock lifts today's variance.
- So turbulence follows turbulence.
- That single link creates clustering.

Calm, Then a Crash

The recursion on two shocks

$$\begin{aligned} \text{calm shock } 0.5 : \sigma^2 &= 0.5 + 0.6(0.25) = 0.65 \text{ (vol 0.81)} \\ \text{big shock } -3.0 : \sigma^2 &= 0.5 + 0.6(9.0) = \mathbf{5.90} \text{ (vol } \mathbf{2.43}) \end{aligned}$$

A calm shock of 0.5 leaves the variance near baseline at 0.65. But a crash-sized shock of -3.0 sends it to 5.90 - a volatility of 2.43%, more than double the long-run level. One big move rewrites tomorrow's risk.

The Spike Decays

Back toward the long-run level

next day, shock 0.8 : $\sigma^2 = 0.5 + 0.6(0.64) = 0.88 \Rightarrow$ the spike decays back toward 1.25

What happens next

- The day after the crash, a normal shock returns.
- The variance falls to 0.88.
- It's already settling down.

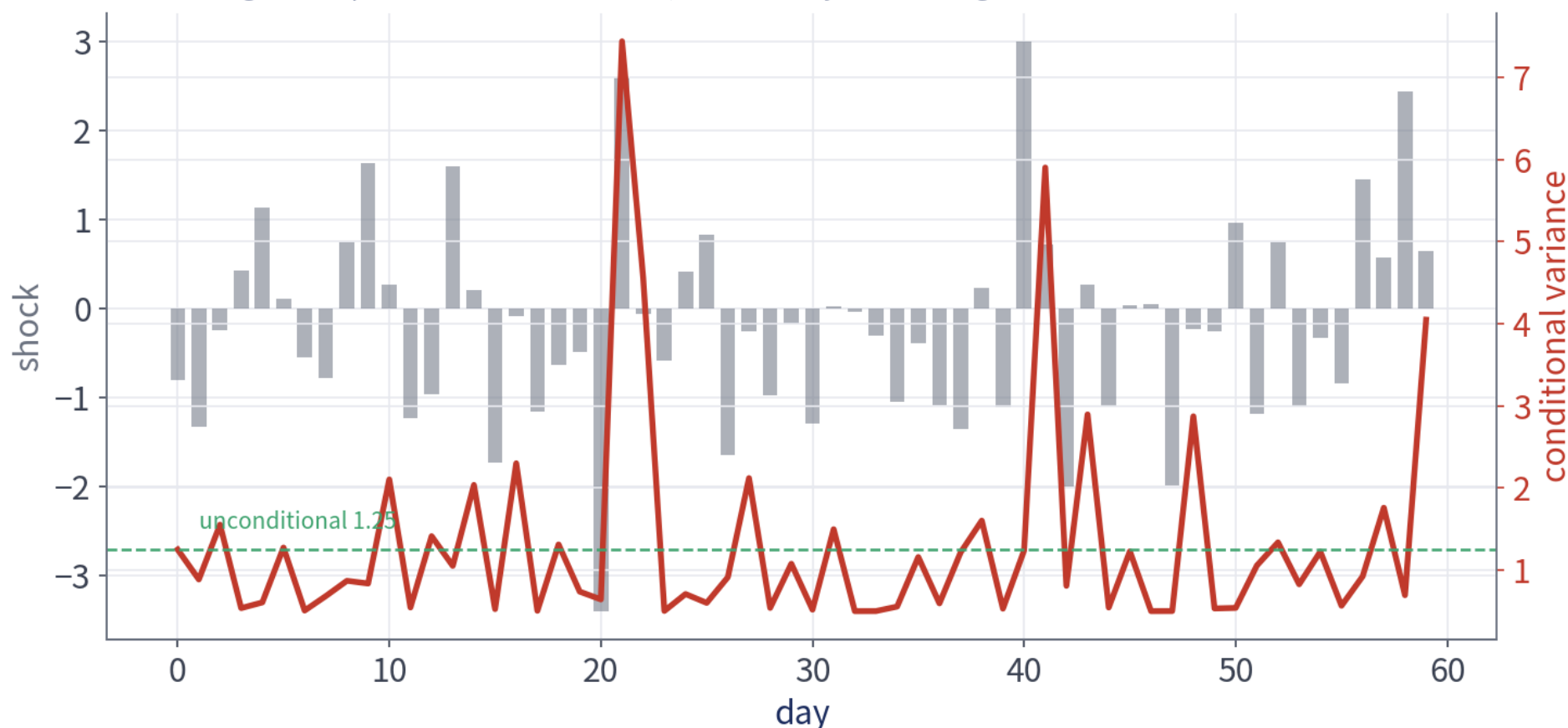
The pattern

- A spike, then a gradual return.
- Toward the unconditional 1.25.
- Clusters form, then dissolve.

Turbulence Follows Turbulence

The grey bars are daily shocks; the red line is the conditional variance. Each large shock lifts the variance sharply, and it decays back toward the dashed long-run level of 1.25 - until the next big move. Calm and stormy periods cluster exactly as real markets do.

Case A: a big shock spikes tomorrow's variance, then it decays - clustering



Why Clustering Matters

Risk is not the same every day

What constant variance misses

- It treats every day as equally risky.
- So it underestimates risk after a shock.
- And overestimates it in calm spells.

What ARCH captures

- Risk that rises right after a big move.
- A time-varying variance you can forecast.
- The clustering markets actually show.

Well-Behaved, and Worth Modeling

Stationarity, then the ARCH test

Before trusting an ARCH model, two checks. First, is it stationary - does a long-run variance even exist? That requires the coefficient below one. Second, are there ARCH effects to model at all? A quick LM test on squared residuals answers that. We compute the unconditional variance and run the test by hand

The two checks

- Stationarity: coefficient below one.
- ARCH effects: an LM test on squared residuals.
- Both come before estimation.

What we'll do by hand

- Compute the unconditional variance.
- See what happens if the coefficient exceeds one.
- Run the LM test statistic.

Unconditional Variance

The long-run level, and when it exists

$$\bar{\sigma}^2 = \frac{\omega}{1-\alpha} = \frac{0.5}{1-0.6} = \mathbf{1.25} \quad (\text{stationary iff } \alpha < 1)$$

The formula

- Baseline over one minus the coefficient.
- It's the level the variance reverts to.
- The dashed line in the last chart.

The condition

- It only makes sense if the coefficient is below one.
- At one or above, the variance has no finite mean.
- The model would explode.

Stationary or Explosive

The coefficient decides

Stationary: $\alpha = 0.6$

- Unconditional variance 0.5 over 0.4.
- Equals 1.25 - finite and well-behaved.
- Shocks fade, variance mean-reverts.

Explosive: $\alpha = 1.2$

- 0.5 over one minus 1.2 is negative.
- No finite long-run variance exists.
- Each shock amplifies - the model blows up.

Stationarity is the multivariate-free version of the unit-root condition: the coefficient must stay below one for the variance to settle rather than explode.

Testing for ARCH Effects

Is there volatility to model?

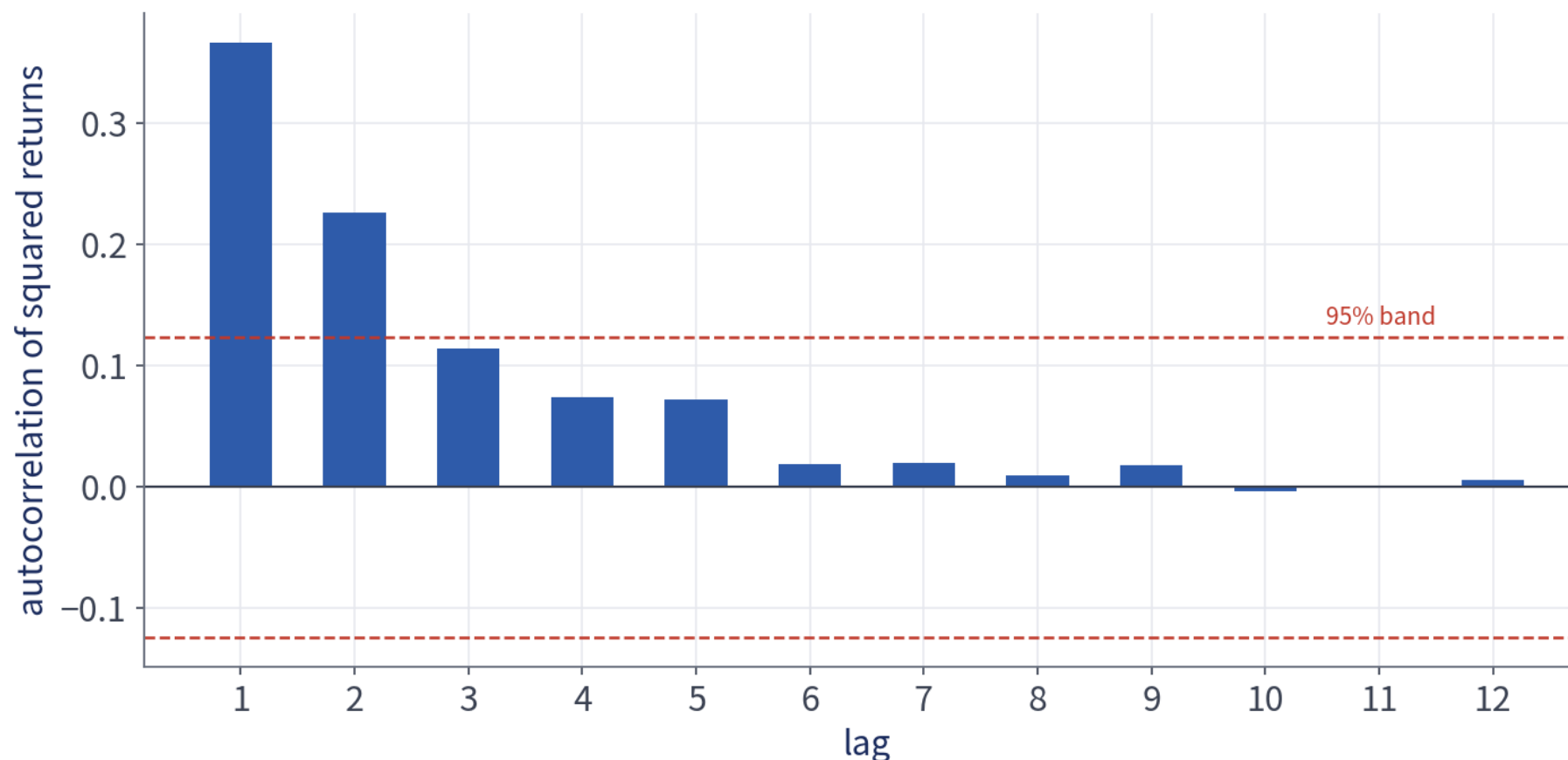
$$LM = T \cdot R^2 = 250 \times 0.08 = \mathbf{20.0} > \underbrace{3.84}_{\chi^2_{1, 5\%}} \Rightarrow \text{ARCH effects present}$$

Regress the squared residuals on their lag and scale the fit: the LM statistic is the sample size times R-squared, here 250 times 0.08, or 20.0. That far exceeds the chi-square cutoff of 3.84 - ARCH effects are present.

The Signature of ARCH

Returns themselves show little autocorrelation, but their squares do - a slowly decaying set of positive bars that pierce the significance band. That persistence in squared returns is exactly what the LM test detects, and what tells you there is volatility clustering to model

Case B: squared returns are autocorrelated - the signature of ARCH effects



Check Before You Model

Two guards on the ARCH fit

Stationarity guards the forecast

- A coefficient below one keeps variance finite.
- The forecast reverts to a stable level.
- Above one, the model is useless.

The test guards your effort

- No ARCH effects means nothing to model.
- The LM test confirms clustering is real.
- Only then is an ARCH model worth fitting.

The ARCH Routine

Test, estimate, check, forecast.

The workflow is short. Test the squared residuals for ARCH effects; if they're there, estimate the recursion; confirm the coefficient is below one for stationarity; then forecast the variance forward. The one link - today's variance from yesterday's squared shock - turns isolated shocks into the clustered volatility markets actually

The recursion (Case A)

- Calm shock: variance 0.65.
- Crash shock: variance 5.90.
- Then it decays toward 1.25.

pl

The checks (Case B)

- Stationary at alpha 0.6 (variance 1.25).
- LM test 20.0 beats 3.84.
- ARCH effects confirmed.

Reproduce the Recursion and the Test

Direct Claude Code to check your arithmetic

GOAL Run the ARCH(1) recursion, check stationarity, and test for ARCH effects.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that, with
omega=0.5, alpha=0.6:
CASE A - recursion:
  * for shocks [0.5, -3.0, 0.8, -0.4], print each
    conditional variance and its volatility.
CASE B - checks:
  * print the unconditional variance;
  * show alpha=1.2 gives no finite variance;
  * with T=250, R^2=0.08, print the LM statistic
    and compare to 3.84.
Then plot the shocks and the conditional variance.
```

Check against the slides

- calm 0.65, crash 5.90.
- unconditional variance 1.25.
- alpha=1.2 explodes.
- $LM = 20.0 > 3.84$.

Deliverable: part1_practice.py whose printed numbers match the recursion and the tests.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What drives the ARCH(1) conditional variance?

A fraction of yesterday's squared shock, plus a baseline

Q2 When is an ARCH(1) model stationary?

When the coefficient alpha is below one

Q3 How do you test for ARCH effects?

An LM test: T times R-squared on squared residuals

All three come from the recursion, the unconditional variance, and the LM test.

Three Pitfalls to Avoid

Where ARCH trips people

Common mistakes

- "Volatility is roughly constant day to day."
- "Fit ARCH without checking the coefficient."
- "Model volatility without testing for it."

The reality

- Risk clusters - it spikes after big moves.
- A coefficient at or above one makes it explode.
- Run the LM test first; there may be nothing to model.

Two Calculations to Remember

Part 1 applications in one screen

1

One squared shock rewrites tomorrow's risk

The ARCH(1) recursion took a calm shock of 0.5 to a variance of 0.65, but a crash-sized -3.0 to 5.90 - a 2.43% volatility, more than double the long-run level - which then decayed back. That is volatility clustering, made concrete.

2

Check stationarity, and test for effects

With a coefficient of 0.6, the unconditional variance is a finite 1.25; at 1.2 it would explode. And an LM statistic of 20.0 - the sample size times R-squared - far exceeded the 3.84 cutoff, confirming ARCH effects are real.

3

A short, disciplined routine

Test the squared residuals, estimate the recursion, confirm stationarity, then forecast the variance. One link between shock and next-day risk is all it takes to model clustering.

Next — Part 2 practice: GARCH, which adds a memory of yesterday's variance for far more parsimonious volatility.

★ WEEK 5 · PART 2 of 3

GARCH and the Volatility Toolkit

The Workhorse That Runs the World's Risk Desks

GARCH(1,1) · persistence & mean reversion · fat tails · the leverage effect

ARCH captured clustering but needed too many lags. GARCH fixes that with one idea — a memory of past variance — and, with two extensions, matches every stylized fact of returns.

2

PART 2 • GARCH

The Volatility Workhorse

GARCH's variance memory, the persistence that governs mean reversion, volatility forecasting, and the extensions for fat tails and the leverage effect.

One Idea Away From the Workhorse

ARCH pointed the way; GARCH completes it

1

ARCH captured clustering

Tying variance to recent squared shocks reproduced clustering and fat tails — the first real model of time-varying risk.

2

But it was inefficient

Volatility's long memory forced a high ARCH order and many parameters — noisy, hard to estimate, prone to over-fitting.

3

GARCH adds memory of past variance

One extra term — yesterday's variance — captures that long memory compactly. GARCH(1,1) does what ARCH(10+) struggled to.

Part 2 in one line: add a variance-memory term, and three parameters model the world's risk.

The GARCH(p,q) Model

Past shocks AND past variance

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad \text{GARCH}(p, q)$$

What is new

- The variance now also depends on its own recent past.
- The β terms carry variance forward day after day.
- p counts variance lags; q counts squared-shock lags.

Why it helps

- Persistence is captured by β , not by piling up ARCH lags.
- Far fewer parameters for the same long memory.
- Almost always, $p = q = 1$ is all you need.

Why Add a Memory of Past Variance?

One term replaces many ARCH lags

A GARCH(1,1) is equivalent to an ARCH with infinitely many lags — the single β term compresses all of that long memory into one parameter.

The problem it solves

- Volatility clusters persist for weeks, not days.
- Pure ARCH needs one lag per day of memory.
- That is wasteful and statistically fragile.

The elegant fix

- $\beta \cdot$ (yesterday's variance) already contains all earlier variance.
- So one recursive term reaches arbitrarily far back.
- Compact, stable, and easy to estimate.

GARCH(1,1): The Workhorse

Three parameters that run the world's risk desks

$$\sigma_t^2 = \underbrace{\omega}_{\text{baseline}} + \underbrace{\alpha \varepsilon_{t-1}^2}_{\text{reaction}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{persistence}} \quad \text{GARCH}(1,1)$$

The three ingredients

- ω : a baseline that anchors the long-run variance level.
- α : the reaction to the most recent shock.
- β : the persistence — how much of yesterday's variance carries over.

Why it dominates

- It fits an enormous range of assets with just three numbers.
- It is the industry default for VaR and option pricing.
- Everything fancier is a variation on this core.

The GARCH(1,1) Recipe

Today's variance is a baseline (ω), plus a reaction to the last shock ($\alpha \cdot \text{shock}^2$), plus a memory of yesterday's variance ($\beta \cdot \text{prev}$). Three ingredients, one recursive rule.

The GARCH(1,1) Recipe

Today's variance = a baseline + a reaction to the last shock + a memory of yesterday's variance.

$\sigma^2_t =$

omega

baseline variance
(the long-run anchor)

+

alpha $e^2_{(t-1)}$

reaction to last shock
(short-run, twitchy)

+

beta $s^2_{(t-1)}$

yesterday's variance
(memory, persistence)

Why beta is the breakthrough

- beta carries variance forward day after day
- one term captures LONG memory that ARCH needs many lags to reproduce
- GARCH(1,1) ~ an infinite ARCH, compactly

The three-parameter workhorse

- omega, alpha, beta: just three numbers
- alpha + beta = persistence (near 1 in practice)
- it runs risk desks, VaR, and option models
- the default first choice for any return series

Persistence and Mean Reversion

What $\alpha + \beta$ tells you

$$\text{persistence} = \alpha + \beta, \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta} \quad (\alpha + \beta < 1)$$

Reading $\alpha + \beta$

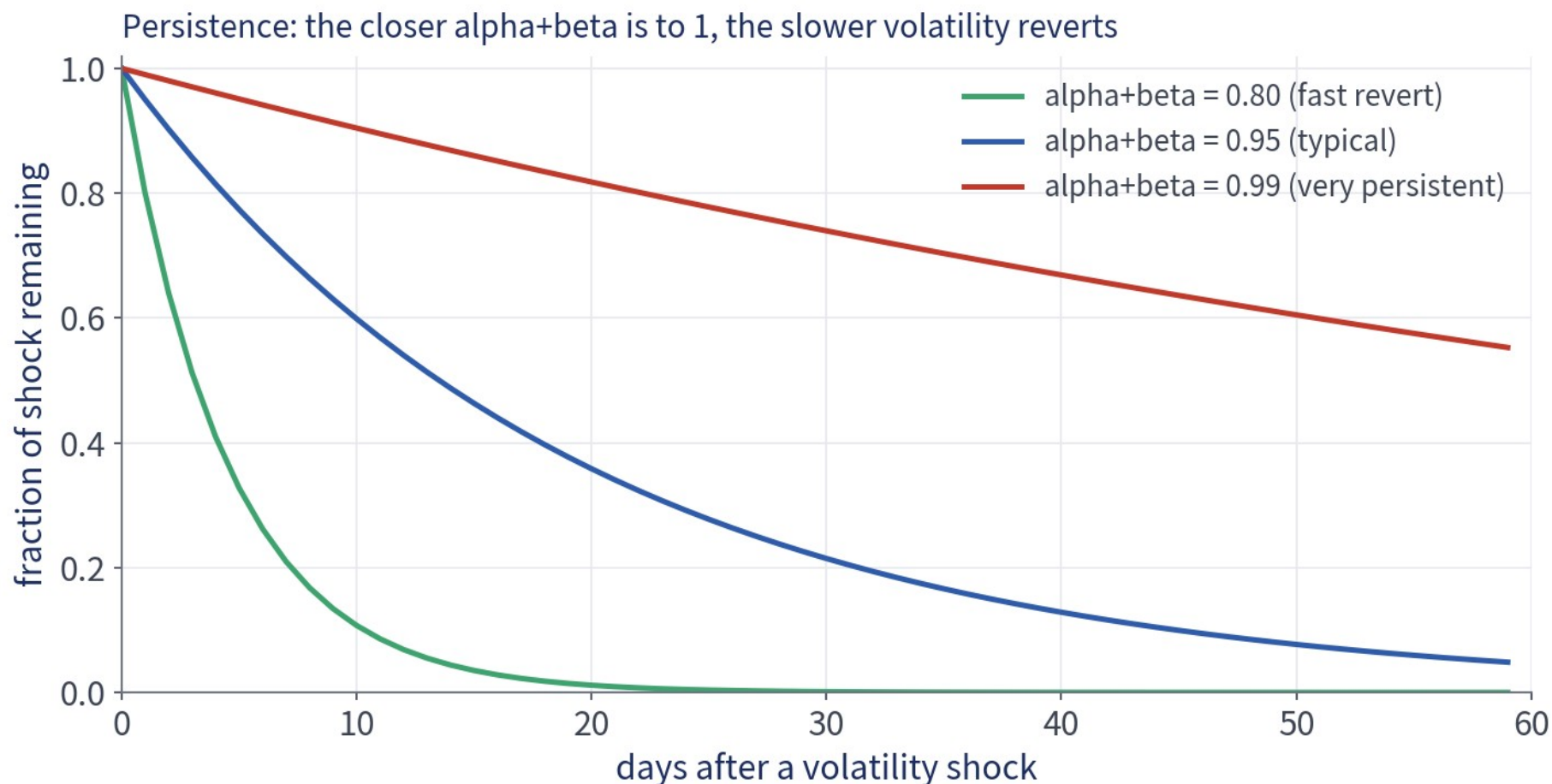
- Their sum is the persistence of a volatility shock.
- Close to 1 (often 0.95–0.99) means very slow reversion.
- The unconditional variance is $\omega/(1 - \alpha - \beta)$.

Why it matters

- It tells you how long a spike in risk will linger.
- It must stay below 1, or variance is not well-defined.
- At exactly 1 (IGARCH), shocks to variance are permanent.

How Fast Does a Volatility Shock Fade?

The closer $\alpha + \beta$ is to 1, the slower a shock to volatility decays. Typical equity estimates near 0.95–0.99 mean risk stays elevated for weeks after a storm.



Volatility Mean-Reverts

Storms and calm both eventually pass

Because $\alpha + \beta$ is below 1, the conditional variance is always pulled back toward its long-run level $\omega/(1 - \alpha - \beta)$. High volatility subsides; unusual calm eventually ends.

What it means

- After a spike, forecast volatility drifts back down over time.
- After deep calm, it drifts back up.
- The long-run level is the anchor it orbits.

Why traders rely on it

- It makes multi-day volatility forecasts sensible.
- It underlies the term structure of implied volatility.
- It is the reason a VIX spike is expected to fade.

The Volatility Half-Life

Turning persistence into a plain-English horizon

How to read it

- Higher persistence → a longer half-life.
- $\alpha + \beta = 0.95$ gives a half-life of roughly two weeks.
- $\alpha + \beta = 0.99$ stretches that to over two months.

Why it is handy

- It communicates risk persistence to non-specialists.
- It sets a sensible horizon for risk limits and hedges.
- It sanity-checks an estimate: is the half-life plausible?

Forecasting Volatility

Tomorrow's risk, and beyond

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \bar{\sigma}^2) \xrightarrow{h \rightarrow \infty} \bar{\sigma}^2$$

One and many steps

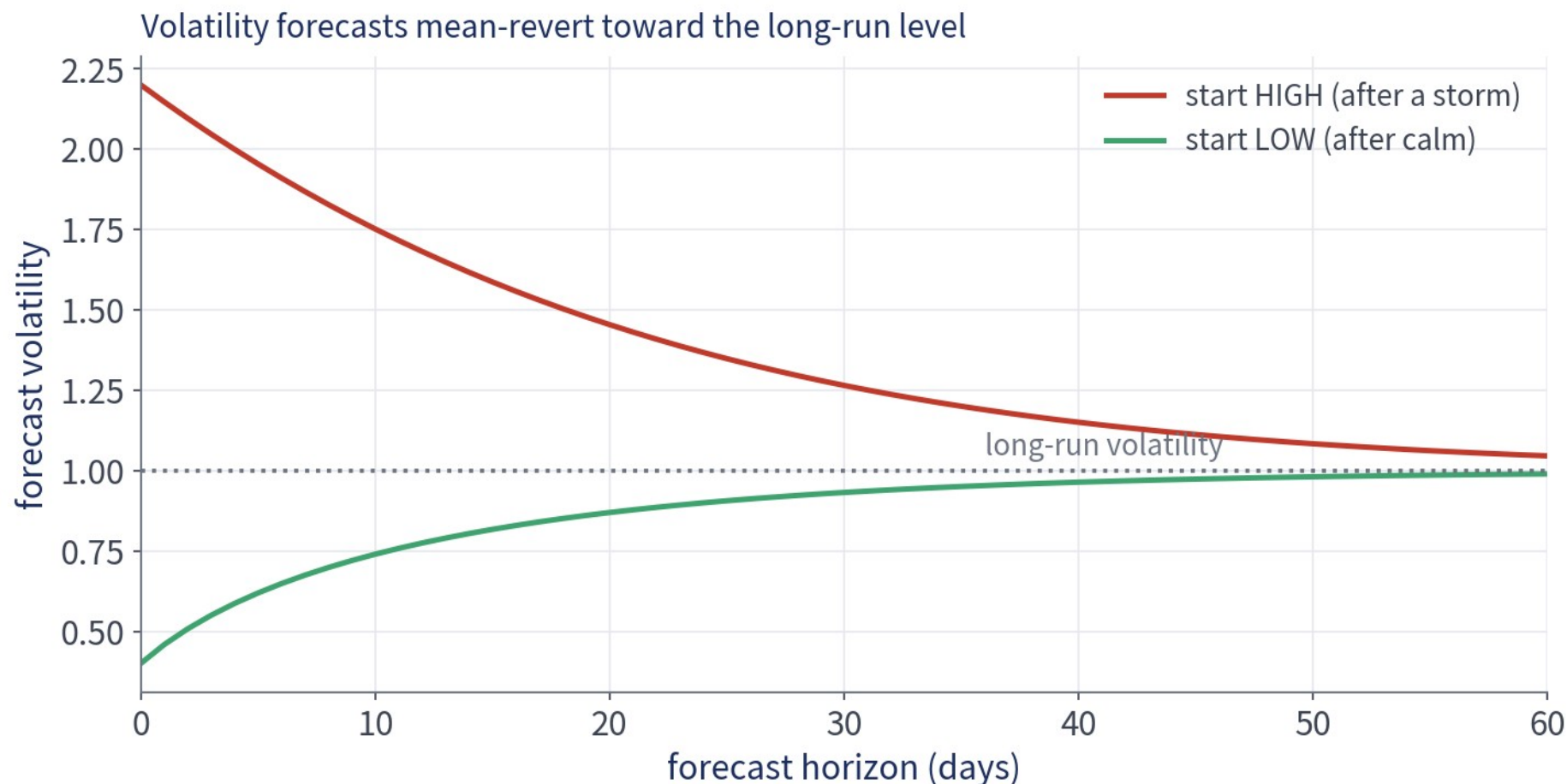
- One step ahead follows directly from the GARCH equation.
- Multi-step forecasts decay geometrically toward the long-run level.
- The speed of decay is governed by $\alpha + \beta$.

The term structure

- Near horizons reflect today's conditions.
- Far horizons converge to the unconditional variance.
- This gives volatility its own 'term structure'.

The Volatility Forecast Term Structure

From a high starting point (after a storm) the forecast decays toward the long-run level; from a low one it rises toward it. Both converge to the same anchor.



Fat Tails: Student-t Innovations

GARCH with normal shocks is not heavy-tailed enough

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t_\nu \text{ (standardized) } \quad \text{heavier tails than normal}$$

The problem

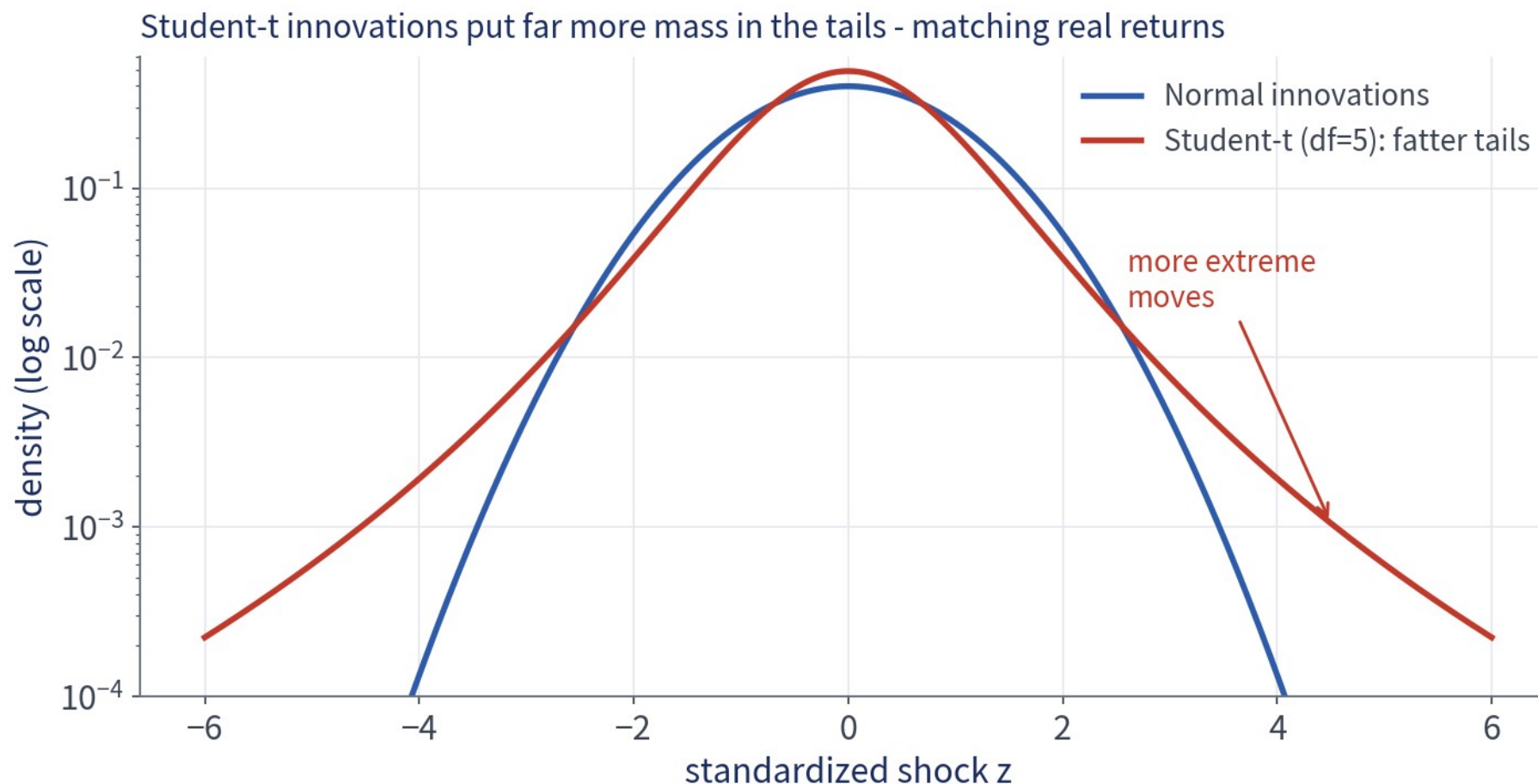
- GARCH already fattens the tails via changing variance.
- But real returns have even heavier tails than that.
- Extreme moves happen more often than normal shocks allow.

The fix

- Draw the standardized shocks z_t from a Student-t.
- The single parameter ν (degrees of freedom) sets tail weight.
- Lower ν means heavier tails — estimated from the data.

Normal vs Student-t Innovations

On a log scale the difference is stark: the Student-t places far more probability in the tails, matching the real frequency of extreme returns.



The Leverage Effect

Plain GARCH treats good and bad news the same

In equity markets, a price drop raises future volatility MORE than an equal-sized rise. Symmetric GARCH — which squares the shock — is blind to this asymmetry.

What we observe

- Bad news (a fall) spikes volatility sharply.
- Good news of the same size raises it far less.
- Squaring the shock erases the sign — and the effect.

Why it is called leverage

- A price fall raises a firm's debt-to-equity ratio.
- Higher leverage makes the equity riskier — more volatile.
- Whatever the cause, the asymmetry is robust and must be modeled.

GJR-GARCH: Asymmetry by an Extra Term

Let bad news add its own kick

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \underbrace{I_{t-1}}_{1 \text{ if } \varepsilon_{t-1} < 0} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad \text{GJR-GARCH}$$

How it works

- An indicator switches on only when the last shock was negative.
- The extra γ term adds volatility just for bad news.
- Good and bad shocks now have different impacts.

Why it is popular

- It is a tiny, intuitive change to GARCH.
- $\gamma > 0$ confirms and measures the leverage effect.
- It is a strong default for equity volatility.

EGARCH: Asymmetry in Logs

Model the LOG of variance

$$\ln \sigma_t^2 = \omega + \alpha g(z_{t-1}) + \beta \ln \sigma_{t-1}^2 \quad \text{EGARCH (asymmetric, always positive)}$$

The idea

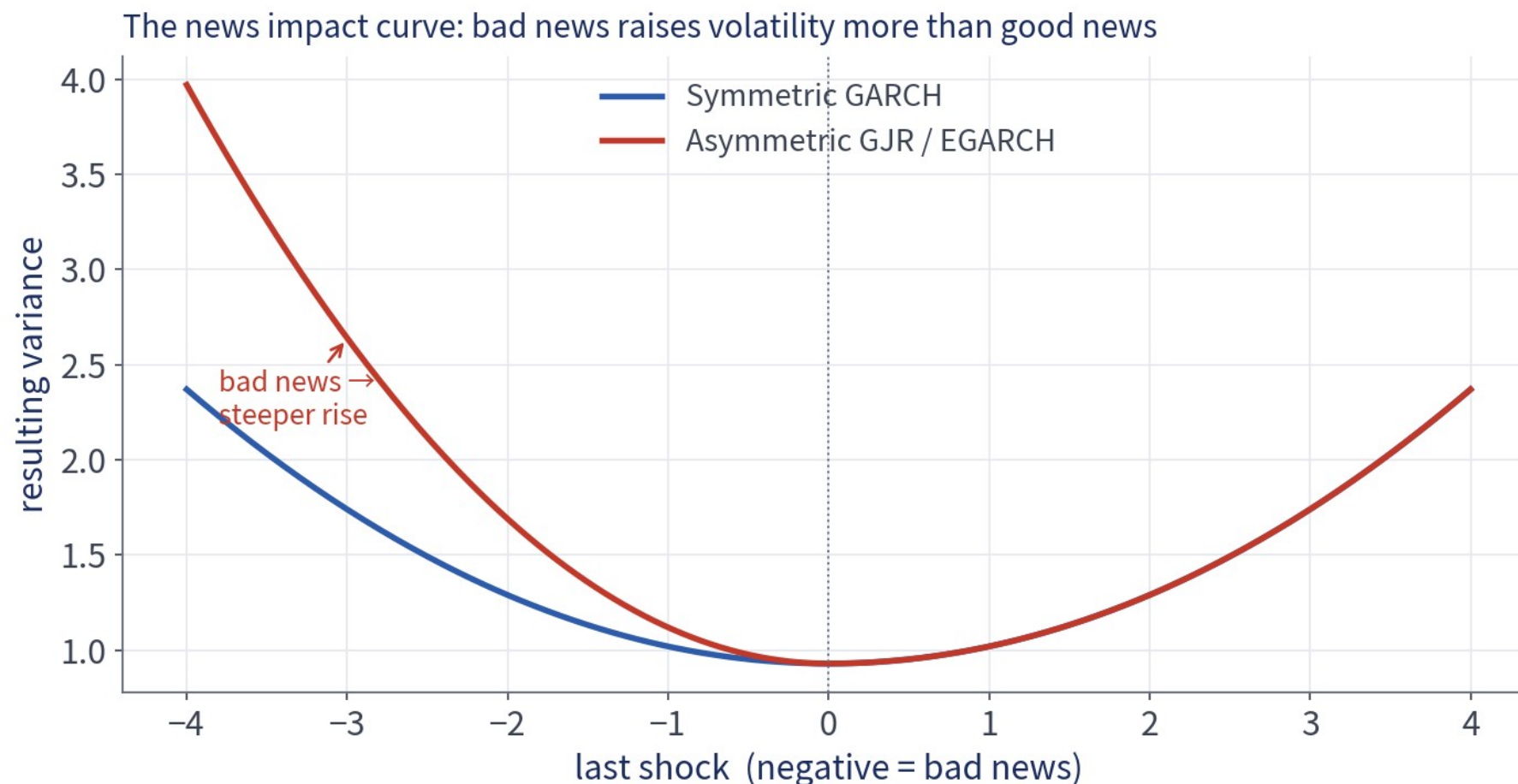
- Model the log of variance, so it is always positive.
- No non-negativity constraints on the parameters.
- Asymmetry enters through the shape of the news function.

GJR vs EGARCH

- Both capture the leverage effect well.
- EGARCH avoids positivity constraints; GJR is simpler to read.
- Try both and let the data (AIC/BIC) choose.

The News Impact Curve

How the last shock maps to today's variance. Symmetric GARCH is a clean parabola; GJR/EGARCH tilt it so bad news (left) raises volatility more than good news.



Choosing a Volatility Model

A short menu, in order of what to add

Model	What it adds	Use when
GARCH(1,1)	variance memory	the baseline — always start here
+ Student-t	heavier tails	extremes are more frequent than normal
GJR-GARCH	leverage / asymmetry	bad news raises volatility more (equities)
EGARCH	asymmetry, always positive	strong leverage; no positivity constraints

A common strong default for equities: a GJR- or E-GARCH(1,1) with Student-t innovations.

Estimating and Checking a GARCH

Fit by likelihood, validate on the residuals

Estimation

- Fit the mean and variance equations by maximum likelihood.
- Choose the innovation distribution (normal or Student-t).
- Confirm $\alpha + \beta < 1$ and all parameters are sensible.

Diagnostics

- Standardized residuals should be $\sim$ iid with no ARCH left.
- Check their squared ACF and re-run the ARCH-LM test.
- Compare candidates (GARCH/GJR/EGARCH) by AIC/BIC.

The Volatility Model Family

Start with GARCH(1,1); add Student-t innovations for fatter tails, and a leverage term (GJR/EGARCH) for asymmetry. Together they cover the stylized facts.

The Volatility Model Family

Start with GARCH, then add heavier tails and asymmetry as the data demands.



Fit the GARCH Family and Compare

Direct Claude Code across GARCH, GJR, and EGARCH

GOAL Fit a GARCH(1,1) and its extensions to real returns, forecast volatility, and let AIC/BIC pick the best.

[INPUT] Prompt · Claude Code

Write and run garch_lab.py that:

1. loads daily equity returns;
2. fits GARCH(1,1), GJR-GARCH, and EGARCH - each with normal AND Student-t innovations - using the 'arch' library;
3. reports alpha+beta (persistence), the half-life, and AIC/BIC for each;
4. plots the fitted conditional volatility and a 20-day volatility forecast for the winner, plus its news-impact curve.

Tell me which model wins and whether leverage and fat tails are significant.

How to work with it

- Expect persistence near 0.95–0.99 for equities.
- GJR/EGARCH usually beat plain GARCH on stocks.
- Student-t usually improves the fit noticeably.
- Keep it; Part 3 turns this into VaR and bands.

Deliverable: garch_lab.py + a model-comparison table and a volatility forecast.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

A fitted GARCH(1,1) has $\alpha + \beta = 0.99$. What does that imply about risk?

Very persistent

Q2

Your standardized residuals still show fat tails. What extension helps?

**Student-t
innovations**

Q3

Bad news raises volatility more than good news. Which models capture this?

GJR / EGARCH

All three come from persistence, the innovation distribution, and the leverage effect.

Three Pitfalls to Avoid

Where GARCH modeling goes wrong

Common mistakes

- "Higher-order GARCH always fits better."
- "Normal innovations are good enough."
- "Symmetric GARCH is fine for equities."

The reality

- GARCH(1,1) almost always suffices; higher orders rarely help.
- Real returns need Student-t tails — normal understates extremes.
- Equities show leverage; use GJR or EGARCH to capture it.

Three Things to Carry Forward

Part 2 in one screen

1

GARCH adds variance memory

One β term captures the long memory that ARCH needed many lags for. GARCH(1,1) — ω, α, β — is the three-parameter workhorse of risk modeling.

2

Persistence governs everything

$\alpha + \beta$ sets how slowly volatility reverts, its half-life, and its forecast term structure — near 1 means storms of risk fade slowly.

3

Two extensions complete the picture

Student-t innovations for fat tails, and GJR/EGARCH for the leverage effect — together they match every stylized fact of returns.

Next — Part 3: putting GARCH to work — Value-at-Risk, back-testing, and pairs-trading bands.

★ WEEK 5 · PART 2 · PRACTICE

The GARCH Model, Worked by Hand

Add a Memory of Variance, Then Forecast the Storm

The GARCH(1,1) recursion, persistence, and the forecast

Part 2's workhorse, with the arithmetic shown. We run the GARCH(1,1) recursion - baseline, reaction, and a memory of yesterday's variance - then read persistence and the volatility half-life, and forecast risk forward as it mean-reverts to its long-run level.

2P

PART 2 · APPLICATIONS

Memory Makes Volatility Persist

Case A: run the GARCH(1,1) recursion, then compute persistence, the long-run variance, and the volatility half-life.
Case B: forecast volatility several steps ahead as it reverts to its long-run level, and meet the leverage and fat-tail extensions. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

GARCH adds a memory of variance

To the ARCH reaction it adds yesterday's variance itself, so one beta term replaces many ARCH lags - far more parsimonious.

2

Alpha plus beta is persistence

Their sum tells you how long volatility shocks last. Near one means highly persistent, with a slow-decaying half-life.

3

Volatility mean-reverts, so it forecasts

The forecast pulls back to the long-run variance at the persistence rate - tomorrow's risk, and the whole horizon beyond.

Everything here is a calculator exercise: one recursion, one half-life, one forecast path.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the three GARCH parameters.
- Run the recursion, or forecast forward.
- Then read persistence and risk.

What you should watch

- How the memory term makes volatility persist.
- How persistence sets the half-life.
- How the forecast reverts to the long run.

The GARCH(1,1) Recursion

Baseline, reaction, and memory

GARCH(1,1) runs the world's risk desks with just three numbers.

Today's variance is a baseline, plus a reaction to yesterday's squared shock, plus a memory of yesterday's variance. We run the recursion once, then read the persistence, the long-run variance, and the volatility half-life it implies

The model

- Baseline omega 0.02.
- Reaction alpha 0.09, memory beta 0.90.
- Yesterday: shock -2.0, variance 1.0.

What we'll do by hand

- Run the recursion for today's variance.
- Add up persistence.
- Compute the long-run level and half-life.

The GARCH(1,1) Recipe

Three terms, one variance

$$\sigma_t^2 = \underbrace{\omega}_{\text{baseline}} + \underbrace{\alpha \varepsilon_{t-1}^2}_{\text{reaction}} + \underbrace{\beta \sigma_{t-1}^2}_{\text{memory of variance}} \quad (\text{one } \beta \text{ replaces many ARCH lags})$$

The three terms

- A constant baseline the variance drifts around.
- A reaction to the latest squared shock.
- A memory of yesterday's variance.

Why the memory helps

- One beta term captures a long shock history.
- It replaces many ARCH lags.
- Three parameters do the work of dozens.

Run the Recursion

Baseline, reaction, memory

$$\sigma_t^2 = 0.02 + 0.09(-2.0)^2 + 0.90(1.0) = \underbrace{0.02}_{\text{base}} + \underbrace{0.36}_{\text{react}} + \underbrace{0.90}_{\text{memory}} = 1.28$$

The three pieces

- Baseline contributes 0.02.
- The reaction to the -2.0 shock adds 0.36.
- The memory carries 0.90 forward.

The total

- They sum to a variance of 1.28.
- The memory term dominates.
- That's what makes volatility persist.

Persistence and Half-Life

How long storms last

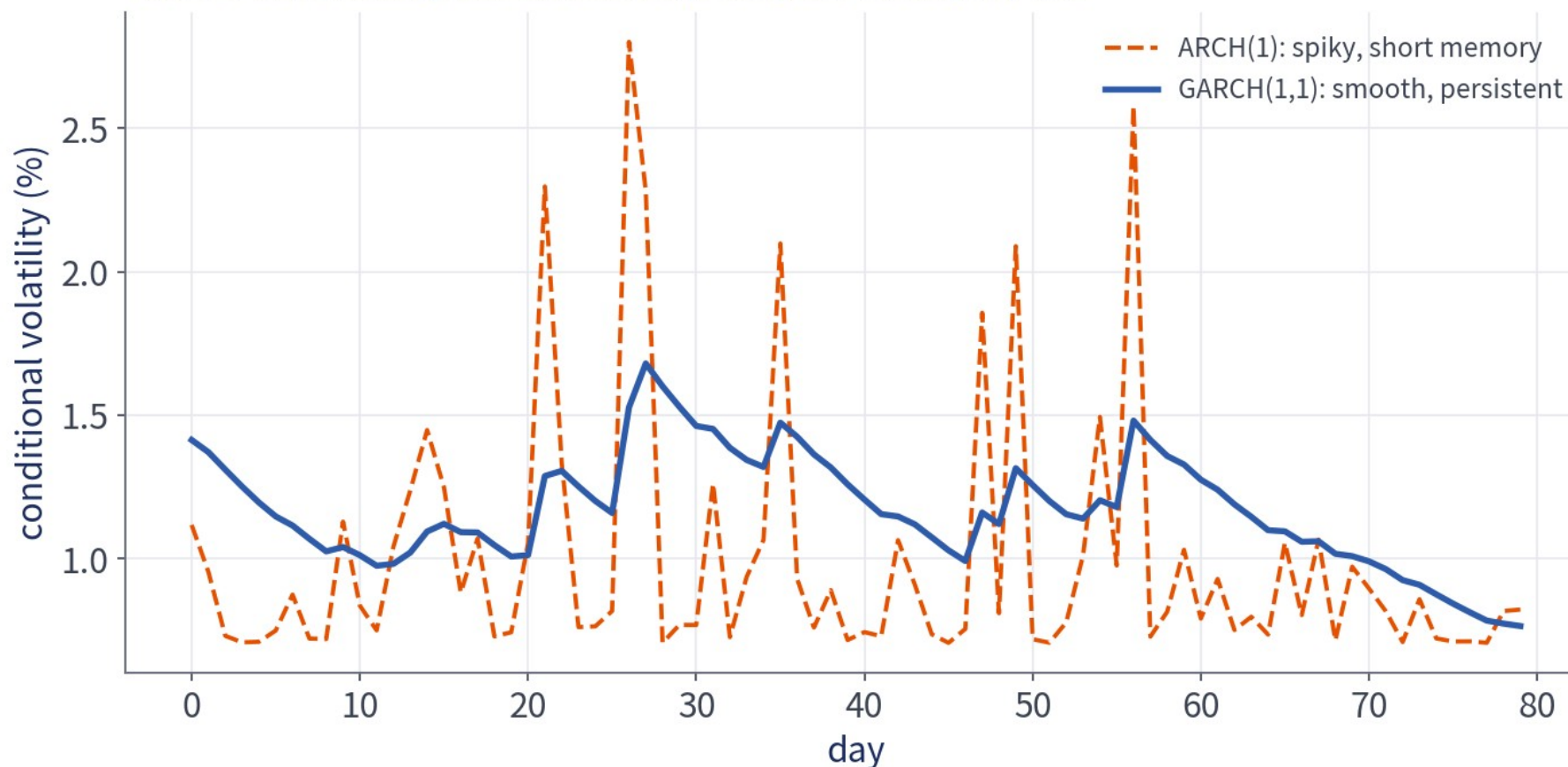
$$\alpha + \beta = 0.99 \text{ (persistence),} \quad \bar{\sigma}^2 = \frac{\omega}{1 - 0.99} = \mathbf{2.0}$$
$$\text{half-life} = \frac{\ln 0.5}{\ln 0.99} = \mathbf{69} \text{ days}$$

Alpha plus beta is 0.99 - very persistent. The long-run variance is 2.0, and the volatility half-life is 69 days: a shock takes about ten weeks to fade halfway. Typical of equity markets.

Memory Smooths the Storm

On the same shocks, ARCH (dashed) spikes and drops sharply - it has no memory. GARCH (solid) rises and then decays gradually, because yesterday's variance carries into today. That persistence is why GARCH matches the slow-fading volatility of real markets with only three parameters

Case A: GARCH's memory term smooths and prolongs the volatility response



Why GARCH Won

Parsimony and persistence

What the memory buys

- A long volatility history in one beta.
- Persistence markets actually display.
- Three parameters instead of many lags.

Why it's the workhorse

- Simple enough to estimate reliably.
- Rich enough to fit equity volatility.
- The default on every risk desk.

Forecasting the Storm

Risk over the whole horizon

Because volatility mean-reverts, GARCH forecasts it. Today's variance sits below its long-run level, so the forecast climbs back toward it at the persistence rate - fast if persistence is low, slow if it's near one. We compute the forecast at several horizons, then meet the leverage and fat-tail extensions that make it realistic

The starting point

- Today's variance is 1.28.
- The long-run level is 2.0.
- Persistence is 0.99 - slow reversion.

What we'll do by hand

- Forecast at one, five, and twenty days.
- Watch it climb toward the long run.
- Then weigh the extensions.

The Multi-Step Forecast

Mean reversion to the long run

$$\mathbb{E}[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^h(\sigma_t^2 - \bar{\sigma}^2) \quad (\text{reverts to the long-run level})$$

How it works

- Start from today's variance.
- Decay the gap to the long run.
- At the persistence rate each step.

What it implies

- Low persistence: fast reversion.
- High persistence: a slow, gentle climb.
- The long run is the anchor.

The Forecast Path

Climbing back to the long run

horizon	variance forecast	volatility
$h = 1$	1.29	1.14%
$h = 5$	1.32	1.15%
$h = 20$	1.41	1.19%
$h = 100$	1.74	1.32%

From today's 1.28, the forecast inches up - 1.29, 1.32, 1.41, 1.74 - toward the long-run 2.0. The climb is slow because persistence is 0.99; even at a hundred days the forecast hasn't fully arrived.

Two Realistic Extensions

Leverage and fat tails

leverage: bad news raises vol more (GJR); fat tails: Student- t innovations, not normal

The leverage effect

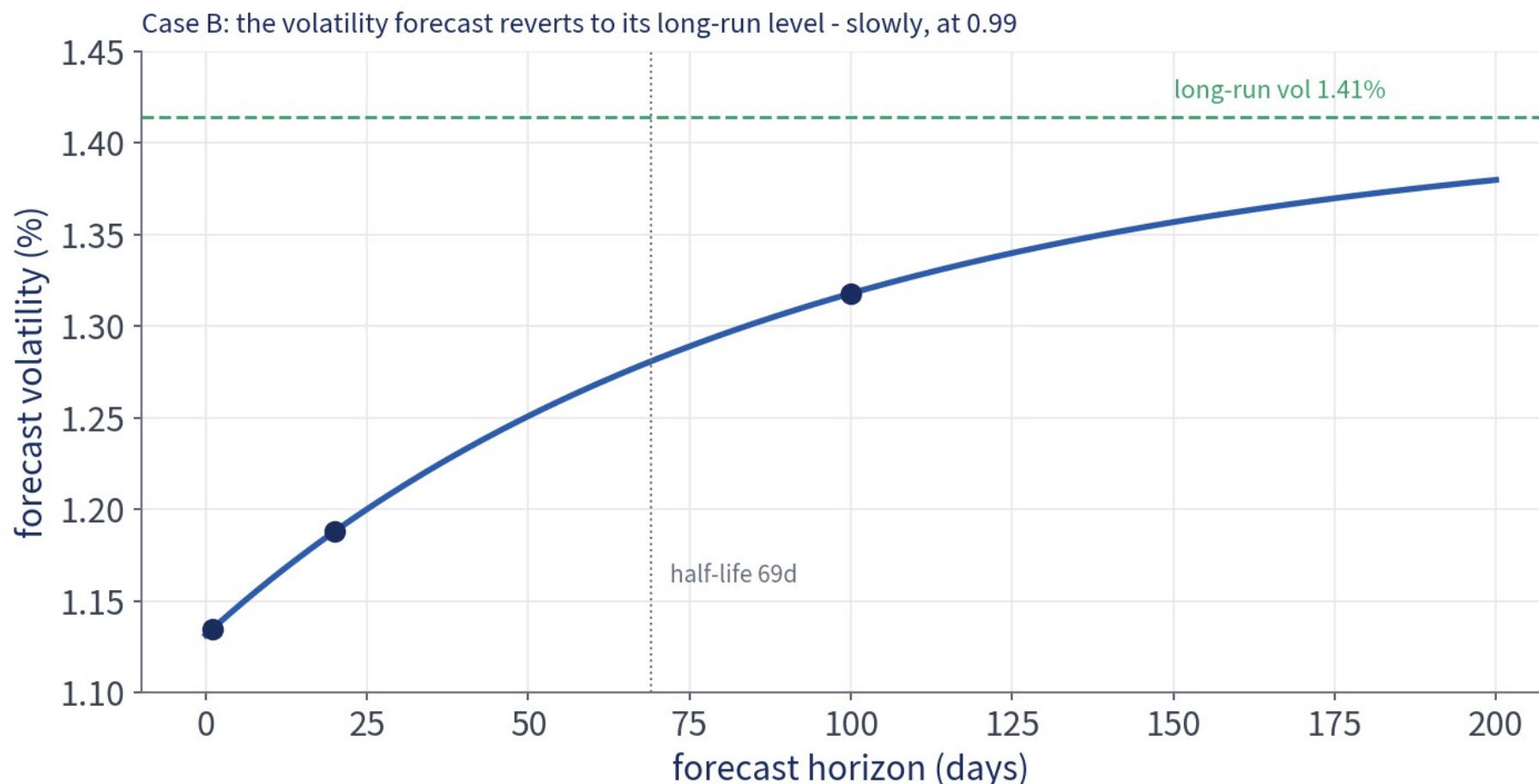
- Bad news raises volatility more than good.
- Plain GARCH treats them the same.
- GJR-GARCH adds an asymmetry term.

Fat tails

- Normal shocks are too thin-tailed for returns.
- Real crashes happen more often than normal says.
- Student- t innovations fix the kurtosis.

The Volatility Term Structure

Starting below its long-run level, the forecast volatility climbs slowly toward the dashed 1.41% line. The half-life of 69 days marks where it has closed half the gap. High persistence makes this a long, gentle curve - today's calm lingers for weeks before fully normalizing



Risk Has a Horizon

And a realistic shape

Why the forecast matters

- Risk over one day differs from over a month.
- The term structure feeds directly into VaR.
- Persistence sets how long calm or storm lasts.

Why the extensions matter

- Leverage captures the crash asymmetry.
- Fat tails price the true chance of extremes.
- Together they make the risk numbers honest.

The GARCH Workhorse

Why it works the risk best

GARCH(1,1) earns its place with three numbers. The baseline sets the long-run level, the reaction responds to the latest shock, and the memory makes volatility persist. Persistence gives the half-life; mean reversion gives the forecast. Add leverage and fat tails, and you have a model that captures how real market risk clusters,

The recursion (Case A)

- Variance 1.28: base + react + memory.
- Persistence 0.99, half-life 69 days.
- Long-run variance 2.0.

isi

The forecast (Case B)

- Climbs 1.29 to 1.74 over 100 days.
- Reverts to the long-run 2.0.
- Leverage and fat tails refine it.

Reproduce the Recursion and the Forecast

Direct Claude Code to check your arithmetic

GOAL Run GARCH(1,1), read persistence and half-life, then forecast the variance.

[INPUT] Prompt · Claude Code

```
Write and run part2_practice.py that, with
omega=0.02, alpha=0.09, beta=0.90:
CASE A - recursion:
    * with shock -2.0 and prev variance 1.0, print
      today's variance broken into its three terms;
    * print persistence, unconditional variance,
      and the volatility half-life.
CASE B - forecast:
    * from variance 1.28, print the h-step forecast
      for h = 1, 5, 20, 100.
Then plot ARCH vs GARCH, and the forecast path.
```

Check against the slides

- variance 1.28
($0.02 + 0.36 + 0.90$).
- persistence 0.99, half-life
69d.
- unconditional variance 2.0.
- forecast 1.29 -> 1.74.

Deliverable: part2_practice.py whose printed numbers match the recursion and the forecast path.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the beta term add over ARCH?

A memory of past variance - one term for many lags

Q2 What does alpha plus beta measure?

Persistence - how long volatility shocks last

Q3 How does GARCH forecast volatility?

It reverts to the long-run level at the persistence rate

All three come from the recursion, the persistence, and the multi-step forecast.

Three Pitfalls to Avoid

Where GARCH trips people

Common mistakes

- "Persistence near one is fine to ignore."
- "Normal shocks capture crash risk."
- "Good and bad news move volatility equally."

The reality

- Near one means very slow reversion - long memory.
- Use Student-t: real tails are fatter than normal.
- Bad news hits harder - use a leverage term.

Two Calculations to Remember

Part 2 applications in one screen

1

Three terms build the variance

The GARCH(1,1) recursion took a baseline of 0.02, a reaction of 0.36 to the -2.0 shock, and a memory of 0.90 to a variance of 1.28. Persistence was 0.99, the long-run variance 2.0, and the volatility half-life 69 days - a slow-fading storm.

2

Volatility mean-reverts, so it forecasts

From today's 1.28, the forecast climbed to 1.29, 1.32, 1.41, and 1.74 over one to a hundred days, reverting toward the long-run 2.0. Leverage and fat-tail extensions then make the shape and the extremes realistic.

3

Three numbers run the risk desks

Baseline, reaction, and memory - plus persistence for the half-life and mean reversion for the forecast. That's why GARCH(1,1) is the volatility workhorse of modern finance.

Next — Part 3 practice: turning a volatility forecast into Value-at-Risk and a volatility-timed pairs trade.

★ WEEK 5 · PART 3 of 3

Volatility in Practice

Value-at-Risk, Back-Testing, and Pairs-Trading Bands

GARCH → VaR & Expected Shortfall · violation back-tests · the spread's breathing bands

A volatility forecast is only useful once it drives a decision. Part 3 turns GARCH into a risk number, tests that number honestly, and launches the pairs-trading thread that runs through the course.

3

PART 3 · IN PRACTICE

Turning Volatility Into Decisions

From a GARCH forecast to Value-at-Risk and Expected Shortfall, back-testing those numbers, and using conditional volatility to build a pairs-trading strategy.

From a Forecast to a Decision

Part 2 built the model; Part 3 puts it to work

1

We can forecast conditional volatility

A fitted GARCH gives today's risk and its forecast path — the size of tomorrow's moves, even though their direction is unknowable.

2

Now we convert risk into numbers and rules

That single volatility feeds Value-at-Risk, Expected Shortfall, position sizing, and trading bands — the everyday tools of a risk desk.

3

And we start the pairs-trading thread

Conditional volatility on a spread gives 'breathing' bands — Stage I of the pairs strategy that recurs through the rest of the course.

Part 3 in one line: a good volatility forecast powers VaR, back-tests, and trading bands.

The GARCH Workflow

From returns to a validated volatility model

1 • Mean

keep simple

Fit a minimal mean model; work with the residuals.

2 • Test

ARCH-LM

Confirm the residuals show volatility clustering.

3 • Fit

GARCH family

Estimate GARCH/GJR/EG ARCH, normal or Student-t.

4 • Validate

residuals

Standardized residuals: no ARCH, right tails.

5 • Forecast

and use

Forecast volatility → VaR, ES, and bands.

Steps 1–4 are Part 2; step 5 is where this part begins — turning the forecast into decisions.

From Volatility to a Risk Number

Making the forecast actionable

A conditional volatility is abstract; a dollar loss limit is not. Value-at-Risk converts today's forecast volatility into a concrete number a risk manager can act on.

What managers need

- A single figure: how bad could a normal-day loss get?
- One that reflects TODAY's risk, not a long-run average.
- One that regulators and desks can monitor daily.

How GARCH supplies it

- It forecasts the size of tomorrow's move directly.
- Scale that by a quantile and you have a loss limit.
- The limit tightens in calm and widens in storms.

Value-at-Risk (VaR)

The loss you should not exceed on a normal day

VaR at 99% is the loss level that should be breached on only 1 day in 100. It is a quantile of the loss distribution — and its width is driven by volatility.

How to read it

- '99% 1-day VaR = 2%' means: worse than -2% on ~1 day in 100.
- A confidence level (95%, 99%) and a horizon (1 day) define it.
- It is a threshold, not the worst case.

Its famous flaw

- VaR says nothing about HOW bad the breach days are.
- Two portfolios with equal VaR can have very different tails.
- Expected Shortfall fixes exactly this (a few slides on).

Value-at-Risk From a GARCH Forecast

Volatility times a quantile

$$\text{VaR}_t(\alpha) = -(\mu + \sigma_t q_\alpha) \quad q_\alpha = \alpha\text{-quantile of } z_t$$

The pieces

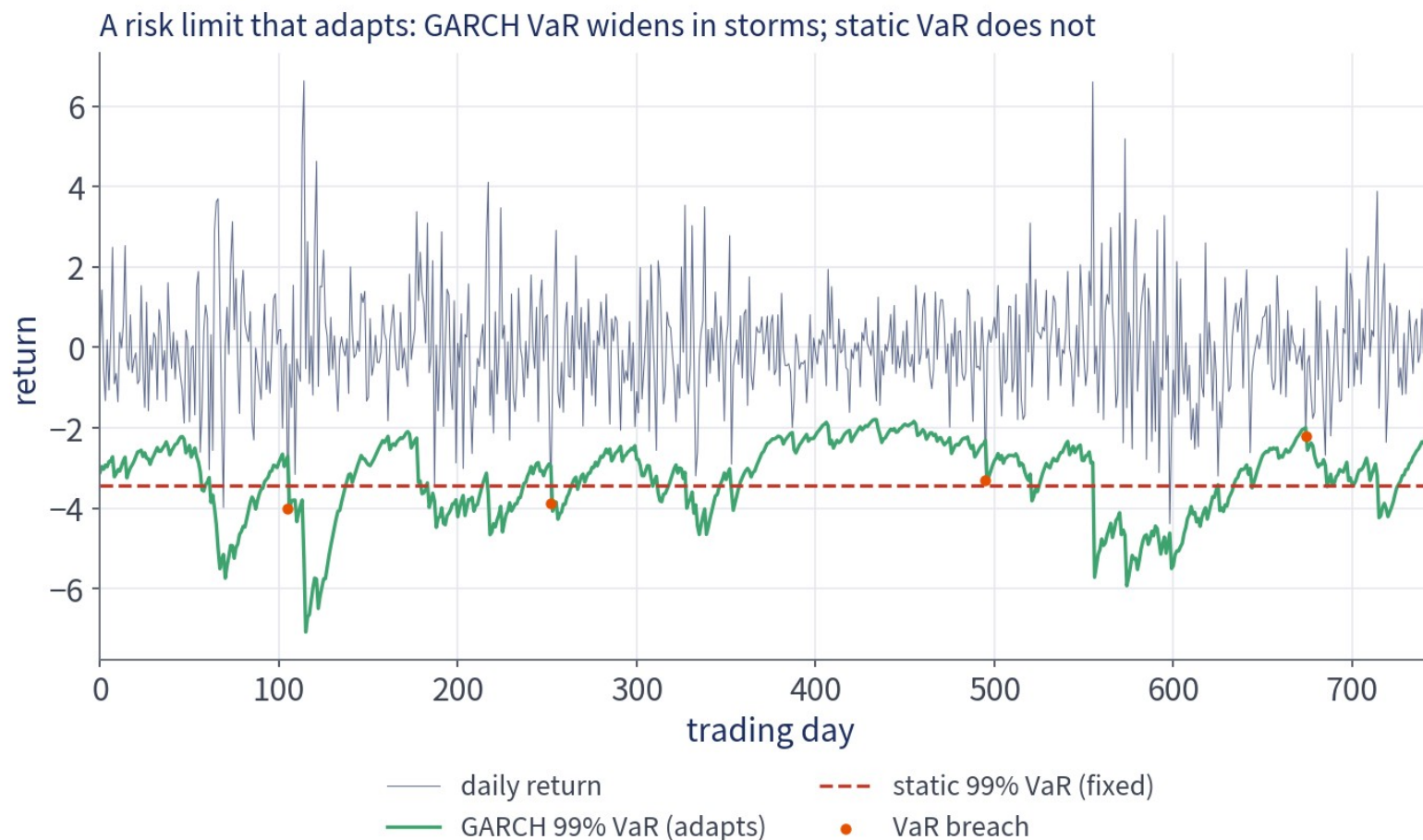
- μ is the (small) expected return; σ_t is today's GARCH volatility.
- q_α is the α -quantile of the standardized innovations.
- Use the Student-t quantile if you fitted fat tails.

Why it adapts

- σ_t changes daily, so the VaR breathes with the market.
- After a shock, σ_t rises and the limit widens automatically.
- This is the whole advantage over a static VaR.

A Risk Limit That Adapts

The GARCH VaR widens in turbulent stretches and tightens in calm ones; a static VaR stays flat and is breached in clusters exactly when risk is highest.



Why a Static VaR Fails

A single number for all seasons is dangerous

The static trap

- A fixed VaR is too loose in calm — it overstates day-to-day risk.
- And far too tight in a crisis — it understates the real danger.
- Its breaches cluster in storms, exactly when it matters most.

The GARCH remedy

- The limit tracks the current regime, day by day.
- Breaches spread out evenly instead of clustering.
- That even spacing is the signature of a well-calibrated VaR.

Expected Shortfall

The average loss WHEN things go wrong

$$ES_t(\alpha) = \mathbb{E} \left[L_t \mid L_t > \text{VaR}_t(\alpha) \right] \quad (\text{expected shortfall})$$

What it adds

- ES is the average loss GIVEN that VaR is breached.
- It looks INTO the tail, not just at its edge.
- It is now the regulatory standard (Basel) for market risk.

Why it is better

- It distinguishes portfolios that VaR rates identically.
- It is coherent — it rewards diversification properly.
- Compute it from the same GARCH forecast and tail quantiles.

Back-Testing VaR

Did reality respect your risk number?

A 99% VaR should be breached about 1% of the time — no more, no less. Back-testing counts the actual breaches and asks whether that rate is statistically right.

How it works

- Compare each day's actual loss to that day's VaR.
- Count the breaches over a long window.
- Compare the count to the expected number, $\alpha \cdot N$.

What to look for

- Too many breaches → the VaR is too optimistic.
- Breaches that CLUSTER → the model misses changing risk.
- Ideally: the right count, evenly spaced.

The Kupiec Test

Is the breach rate statistically correct?

count violations: expect αN ; Kupiec LR test: is the breach rate $= \alpha$?

What it checks

- Under a correct model, breaches occur at rate α .
- Kupiec is a likelihood-ratio test on the observed count.
- A small p-value rejects the VaR model as miscalibrated.

Beyond the count

- Kupiec tests the NUMBER of breaches.
- Independence tests check they do not CLUSTER.
- A good VaR passes both — right count, no clustering.

Back-Testing, Visualized

Cumulative breaches should track the expected 1% line. The GARCH VaR stays close to it; the static VaR drifts above — too many breaches, clustered in storms.



What a Good VaR Model Looks Like

Two things must both be true

The two criteria

- Correct frequency: about α breaches, no systematic bias.
- Independence: breaches do not bunch together in time.
- A model can pass one and fail the other — check both.

The GARCH advantage

- Because it adapts, its breaches are spread through time.
- It reacts to a new regime within a day or two.
- That responsiveness is why it back-tests so well.

Pairs Trading Starts Here

The strategy that runs through the whole course

Recall cointegration (Week 2): two assets whose spread mean-reverts. Trading that reversion needs a threshold — and conditional volatility is what makes the threshold intelligent.

The setup

- Find a cointegrated pair; form its stationary spread.
- Enter when the spread is 'stretched', exit as it reverts.
- But how far is 'stretched'? That depends on the spread's risk.

Where GARCH enters

- Model the spread's conditional volatility with GARCH.
- Set the entry bands at a multiple of that volatility.
- The bands widen in volatile regimes, tighten in calm ones.

The Spread and Its Volatility

Volatility is not constant here either

The spread's dynamics

- The spread mean-reverts (cointegration) but its risk varies.
- Some periods it is quiet; others it swings violently.
- A fixed band ignores this — and fires at the wrong times.

Modeling it

- Apply GARCH to the spread's changes, just as for returns.
- Get a conditional volatility for the spread each day.
- This is the input to the trading bands.

GARCH Conditional Bands on the Spread

Thresholds that scale with the spread's risk

$$S_t = P_t^A - \beta P_t^B, \quad \sigma_{S,t}^2 \text{ modeled by GARCH,} \quad \text{bands} = \pm k \sigma_{S,t}$$

Building the bands

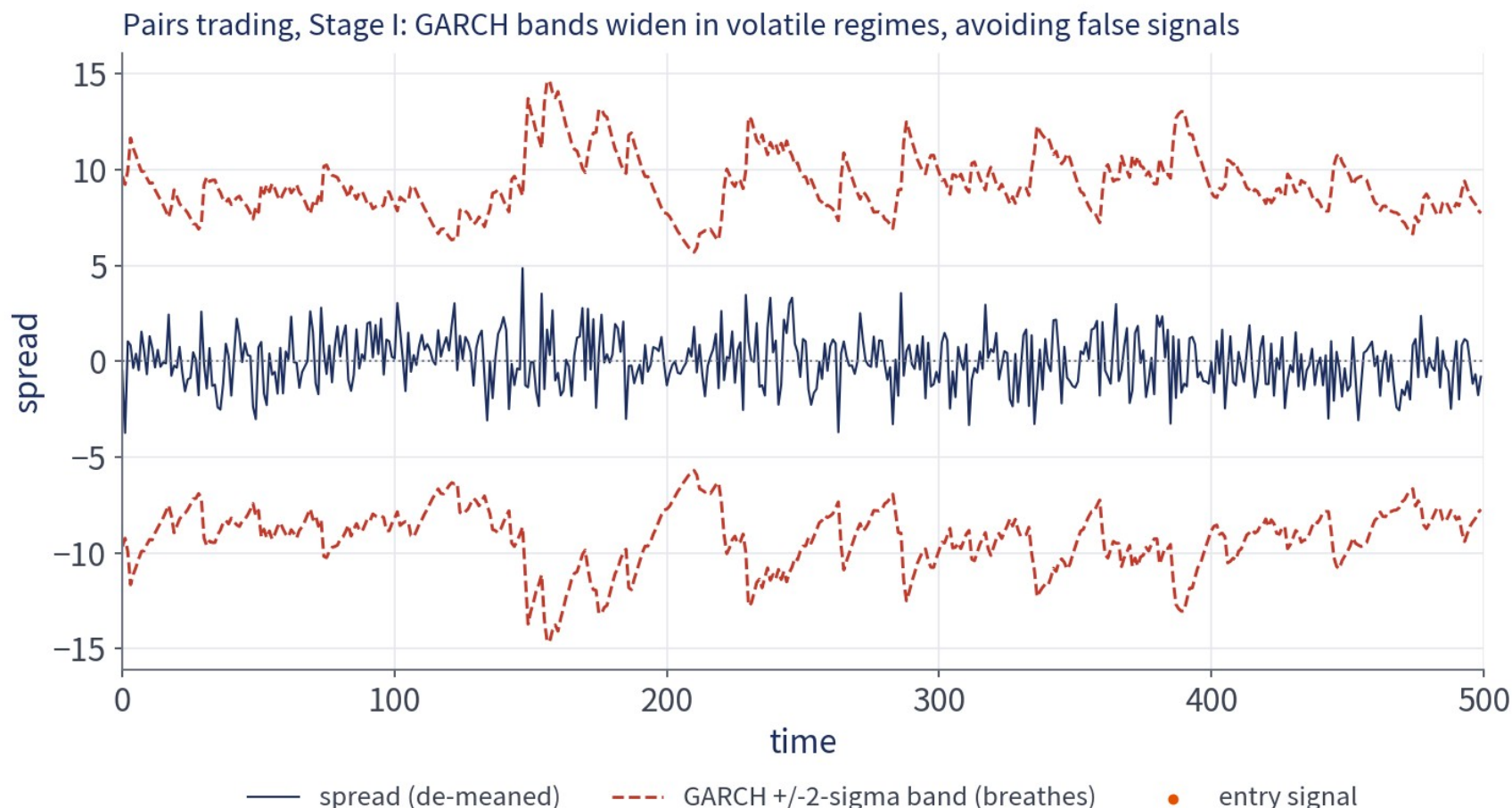
- Form the spread from the cointegrating relationship.
- Model its conditional volatility $\sigma_{S,t}$ with GARCH.
- Set entry/exit bands at $\pm k$ times that volatility.

Why it is smarter

- In calm, tight bands catch small, reliable reversions.
- In turbulence, wide bands avoid premature entries.
- The threshold adapts to the regime automatically.

Pairs Trading, Stage I

The spread with GARCH $\pm 2\sigma$ bands that breathe: they widen in volatile regimes and tighten in calm, so entry signals fire when a move is genuinely unusual.



Why Breathing Bands Beat Fixed Ones

Fewer false signals, better timing

Fixed bands

- Too tight in volatile regimes → constant false entries.
- Too wide in calm regimes → missed reversions.
- They assume a risk level that simply is not constant.

GARCH bands

- Scale the threshold to the spread's current volatility.
- Signals fire only when a move is unusual for TODAY.
- Cleaner entries, and fewer whipsaws in stormy periods.

The Pairs-Trading Roadmap

Where this thread goes from here

Stage I gives us intelligent bands. The later stages — signal generation, position sizing, and portfolio construction — build on this same volatility-aware foundation.

Done here (Stage I)

- Cointegration finds the pair (Week 2).
- GARCH gives the spread's conditional volatility.
- Breathing bands define when the spread is stretched.

Coming later

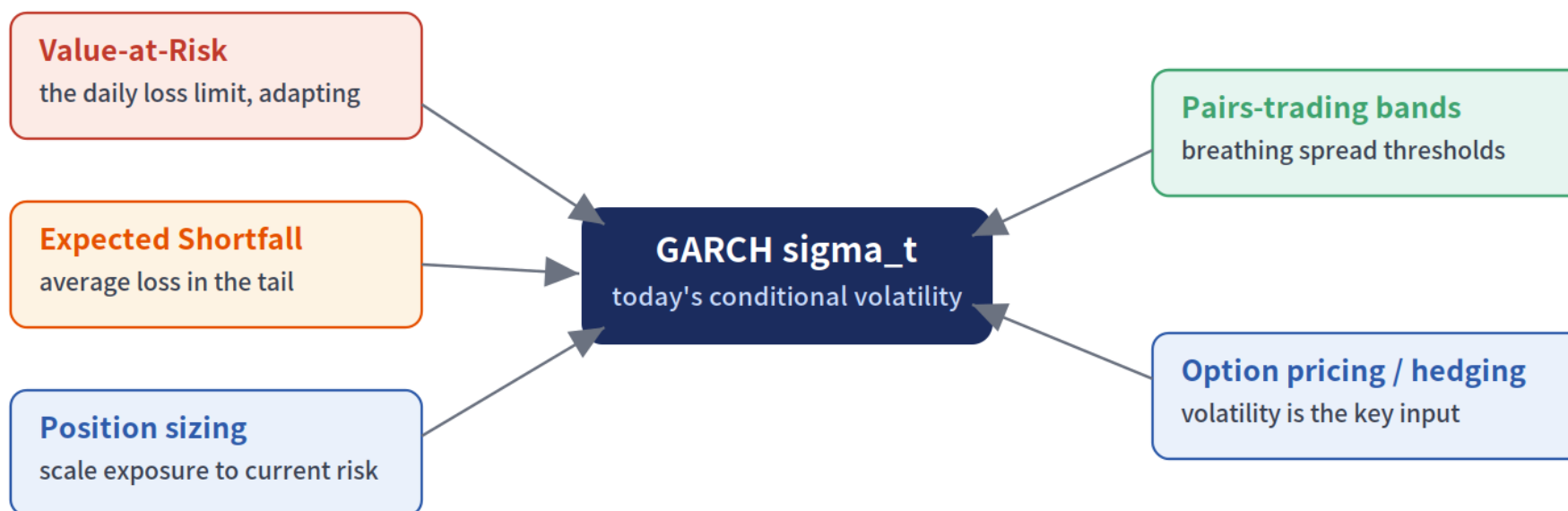
- Turning band crossings into disciplined entry/exit rules.
- Sizing positions by risk, using the same volatility.
- Combining many pairs into a diversified book.

What a Volatility Forecast Unlocks

One GARCH conditional volatility feeds Value-at-Risk, Expected Shortfall, position sizing, option hedging, and pairs-trading bands — the whole risk toolkit.

What a Volatility Forecast Unlocks

One GARCH conditional variance feeds every downstream risk and trading decision.



The theme of the week: you cannot forecast the return, but a good volatility forecast powers nearly every risk decision.

One Asset at a Time Isn't Enough

Why we need to go multivariate

GARCH models each series' volatility alone. But portfolios and pairs depend on how assets move TOGETHER — and correlations, like volatilities, change over time.

The limitation

- Univariate GARCH ignores cross-asset relationships.
- Portfolio risk depends on time-varying correlations.
- Even our pair needs the joint dynamics of two series.

Where the course turns next

- Multivariate models let volatilities and correlations co-move.
- Vector autoregression (VAR) models several series jointly.
- That is exactly where Week 6 begins.

VaR Back-Test and Pairs Bands

Direct Claude Code from risk numbers to a trading signal

GOAL Build a GARCH VaR, back-test it, then form GARCH bands on a cointegrated spread and mark entry signals.

[INPUT] Prompt · Claude Code

Write and run `vol_practice.py` that:

1. fits a $\text{GARCH}(1,1)-t$ to a stock's returns and computes a rolling 99% 1-day VaR;
2. back-tests it: count breaches, run the Kupiec test, and compare to a static VaR;
3. for a cointegrated pair, models the spread's volatility with GARCH and plots ± 2 -sigma bands with entry signals.

Report the breach rate, the Kupiec p-value, and how many signals the breathing bands give versus fixed bands.

How to work with it

- A good VaR: breach rate $\approx 1\%$, Kupiec not rejected.
- Static VaR usually breaches too often, in clusters.
- Breathing bands should give cleaner, fewer signals.
- This closes Week 5 and sets up Week 6.

Deliverable: `vol_practice.py` + a VaR back-test report and a pairs-band chart.

Three Questions Before You Finish

If you can answer these, Week 5 has landed

Q1

Your 99% VaR is breached on 4% of days, in clusters. What's wrong?

VaR too optimistic / static

Q2

Why does Expected Shortfall improve on VaR?

Looks into the tail

Q3

Why do GARCH bands beat fixed bands for a pairs spread?

Adapt to spread volatility

All three follow from time-varying risk, tail measures, and adaptive thresholds.

Three Pitfalls to Avoid

Where volatility-in-practice goes wrong

Common mistakes

- "A single VaR number captures all my risk."
- "If VaR looks fine, no need to back-test."
- "Fixed bands are simpler, so use them for pairs."

The reality

- VaR ignores tail severity — pair it with Expected Shortfall.
- Always back-test: check breach count AND clustering.
- Fixed bands misfire; GARCH bands adapt to the regime.

Week 5 in Three Ideas

Pulling the whole week together

1

Model the variance, not the mean

Part 1: returns are unpredictable in direction but their variance is not. ARCH ties conditional variance to recent squared shocks — clustering, formalized.

2

GARCH is the workhorse

Part 2: a memory term (β) captures long volatility memory in three parameters; Student-t and GJR/EGARCH add fat tails and the leverage effect.

3

Volatility drives every risk decision

Part 3: a GARCH forecast becomes VaR, Expected Shortfall, and breathing pairs-trading bands — validated by back-tests.

Next — Week 6: Multivariate time series and VAR, modeling how assets move together.

★ WEEK 5 · PART 3 · PRACTICE

VaR and Pairs, Worked by Hand

Turn a Volatility Forecast Into a Risk Number, Then a Trade

GARCH Value-at-Risk, the Kupiec test, and conditional bands

Part 3's payoff, with the arithmetic shown. We turn a GARCH volatility forecast into a Value-at-Risk figure and its Expected Shortfall, back-test the number with the Kupiec test, and scale a pairs trade's thresholds with the spread's own conditional volatility.

3P

PART 3 · APPLICATIONS

From a Forecast to a Risk Desk

Case A: convert a GARCH volatility forecast into 95% and 99% Value-at-Risk and the Expected Shortfall beyond it.
Case B: validate the VaR with the Kupiec back-test, and scale a pairs trade's bands with the spread's conditional volatility. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

VaR is volatility times a quantile

A GARCH volatility forecast becomes a loss you shouldn't exceed on a normal day - and it breathes as volatility clusters.

2

Expected Shortfall reads the tail

Beyond the VaR line, ES is the average loss when things do go wrong - always larger, and more honest about extremes.

3

Back-test, then scale the bands

The Kupiec test checks the breach rate, and the spread's own GARCH volatility sets pairs thresholds that widen and tighten.

Everything here is a calculator exercise: one VaR, one shortfall, one back-test, one band.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a GARCH volatility forecast.
- Compute the risk number, or validate it.
- Then act on it.

What you should watch

- How volatility scales into a dollar loss.
- How the Kupiec test judges the model.
- How conditional bands scale with risk.

Value-at-Risk From GARCH

A forecast becomes a risk number

A risk desk needs a single figure: how much could we lose tomorrow on a bad-but-normal day? GARCH gives the volatility forecast; multiplying by a quantile and the position size gives Value-at-Risk. We compute it at 95% and 99%, then the Expected Shortfall - the average loss once that line is breached

The setup

- A \$10M portfolio.
- GARCH forecasts tomorrow's vol at 1.5%.
- A roughly zero daily mean.

What we'll do by hand

- Scale volatility by the 95% and 99% quantiles.
- Multiply by the position.
- Then compute the shortfall.

Value-at-Risk

Volatility times a quantile

$$\text{VaR}_\alpha = z_\alpha \cdot \underbrace{\sigma_t}_{\text{GARCH forecast}} \cdot V \quad (\text{volatility} \times \text{quantile} \times \text{position})$$

The three factors

- The GARCH volatility forecast.
- The quantile of the loss distribution.
- The size of the position.

Why GARCH matters

- The volatility is conditional, not constant.
- So VaR rises after a shock, falls in calm.
- A static number can't do that.

The VaR at Two Levels

95% and 99%

$$95\% : 1.645 \times 0.015 \times \$10\text{M} = \$\mathbf{247\text{K}}$$

$$99\% : 2.326 \times 0.015 \times \$10\text{M} = \$\mathbf{349\text{K}}$$

At 95%, the quantile is 1.645, giving a one-day VaR of about \$247K. At 99%, the quantile rises to 2.326 and the VaR to \$349K. The higher the confidence, the larger the loss you're insuring against.

Expected Shortfall

The average loss in the tail

$$ES_{95\%} = \frac{\phi(z_\alpha)}{\alpha} \sigma_t V = \frac{0.1031}{0.05} \times 0.015 \times \$10M = \$309K (> VaR)$$

What it measures

- Not the threshold, but the average beyond it.
- The mean loss on the worst 5% of days.
- Here about \$309K.

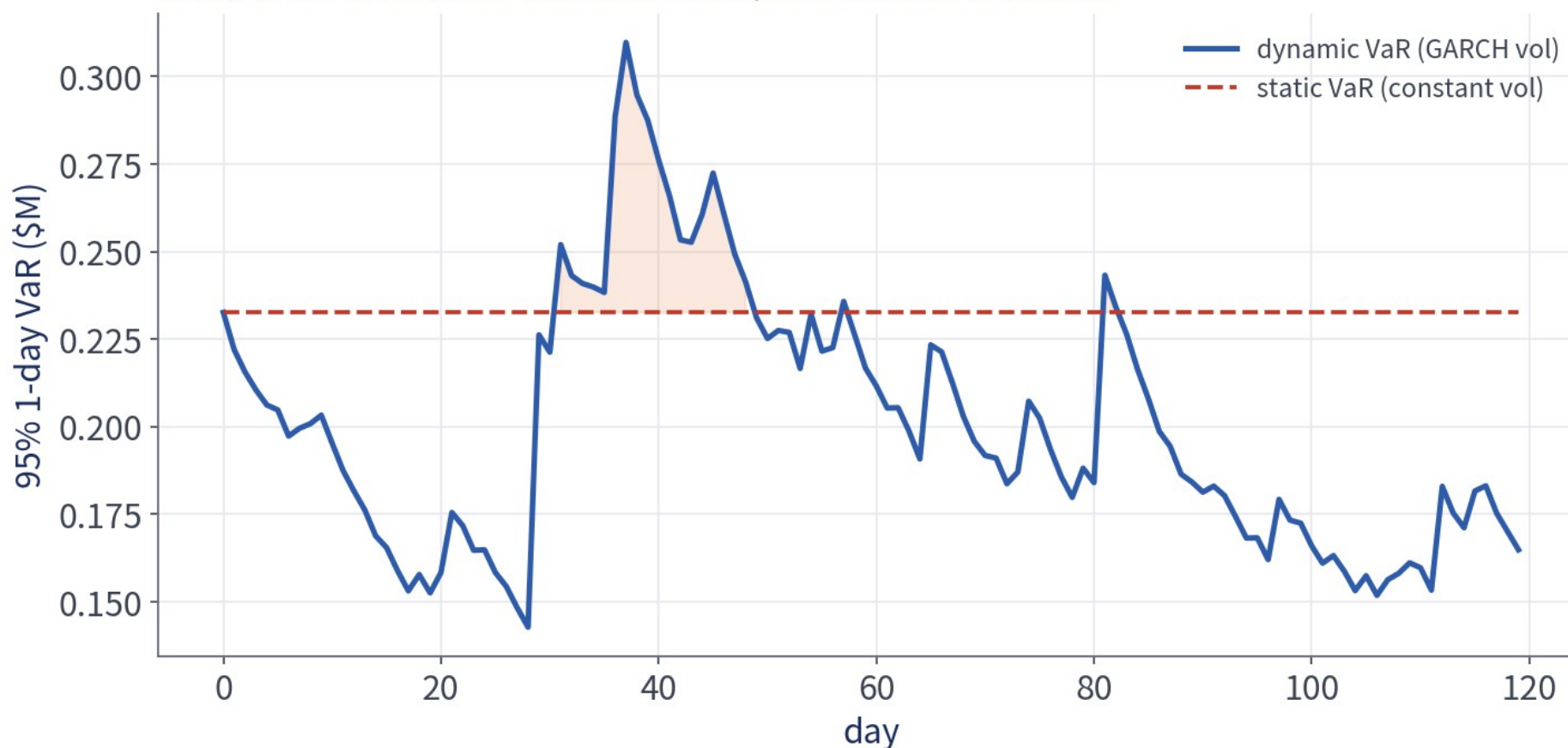
Why it's larger

- VaR is only the edge of the tail.
- ES averages the losses deeper inside it.
- So ES always exceeds VaR.

VaR That Breathes

The GARCH-based VaR (blue) rises sharply after each shock and relaxes in calm spells, tracking the conditional volatility. The static VaR (dashed red) is flat - and in the shaded periods it sits below the dynamic line, understating the true risk exactly when it matters most.

Case A: GARCH VaR breathes - it rises after shocks, when static VaR understates



A Living Risk Number

Why static VaR is dangerous

The static trap

- One VaR for all seasons.
- It understates risk after a shock.
- Breaches cluster right when they hurt.

The GARCH fix

- VaR scales up as volatility clusters.
- It relaxes when markets calm.
- ES then sizes the tail beyond it.

Validate, Then Trade

Back-test the VaR, scale the bands

A risk number is only as good as its track record. The Kupiec test checks whether the VaR was breached at the right rate - too often means the model understates risk. Then we bring GARCH to the pairs trade of Weeks 6 and 7: the spread has its own conditional volatility, so its trading bands should widen and tighten with it

The validation

- Count the VaR breaches over 250 days.
- Compare to the expected 12.5.
- The Kupiec test judges the gap.

What we'll do by hand

- Run Kupiec on a good and a bad model.
- Then build conditional bands on the spread.
- Compare to static thresholds.

The Kupiec Test

Is the breach rate correct?

$$LR = 2 \left[x \ln \frac{x/N}{p} + (N - x) \ln \frac{1-x/N}{1-p} \right] \sim \chi_1^2 \quad (\text{is the breach rate correct?})$$

The idea

- A 95% VaR should be breached about 5% of days.
- Count the actual breaches.
- Test whether the rate is statistically right.

The verdict

- The statistic follows a chi-square.
- Below 3.84, the model passes.
- Above, too many breaches - reject it.

Two Models, Two Verdicts

The test catches the bad one

breaches (of 250)	rate	LR statistic	verdict
14 (expected 12.5)	5.6%	0.18	PASS
25	10.0%	10.33	REJECT

25 model with 14 breaches - close to the expected 10.33 - gives a Kupiec statistic of 0.18, well under 3.84: it passes. But 25 breaches, a 10% rate, pushes the statistic to 10.33 and is firmly rejected - that VaR understates risk.

Conditional Bands on the Spread

Thresholds that scale with risk

spread bands = $\bar{s} \pm k \sigma_{s,t}$ (GARCH) : wide when volatile, tight when calm

The idea

- The spread's volatility clusters too.
- Fixed z-score bands ignore that.
- GARCH bands scale with the spread's risk.

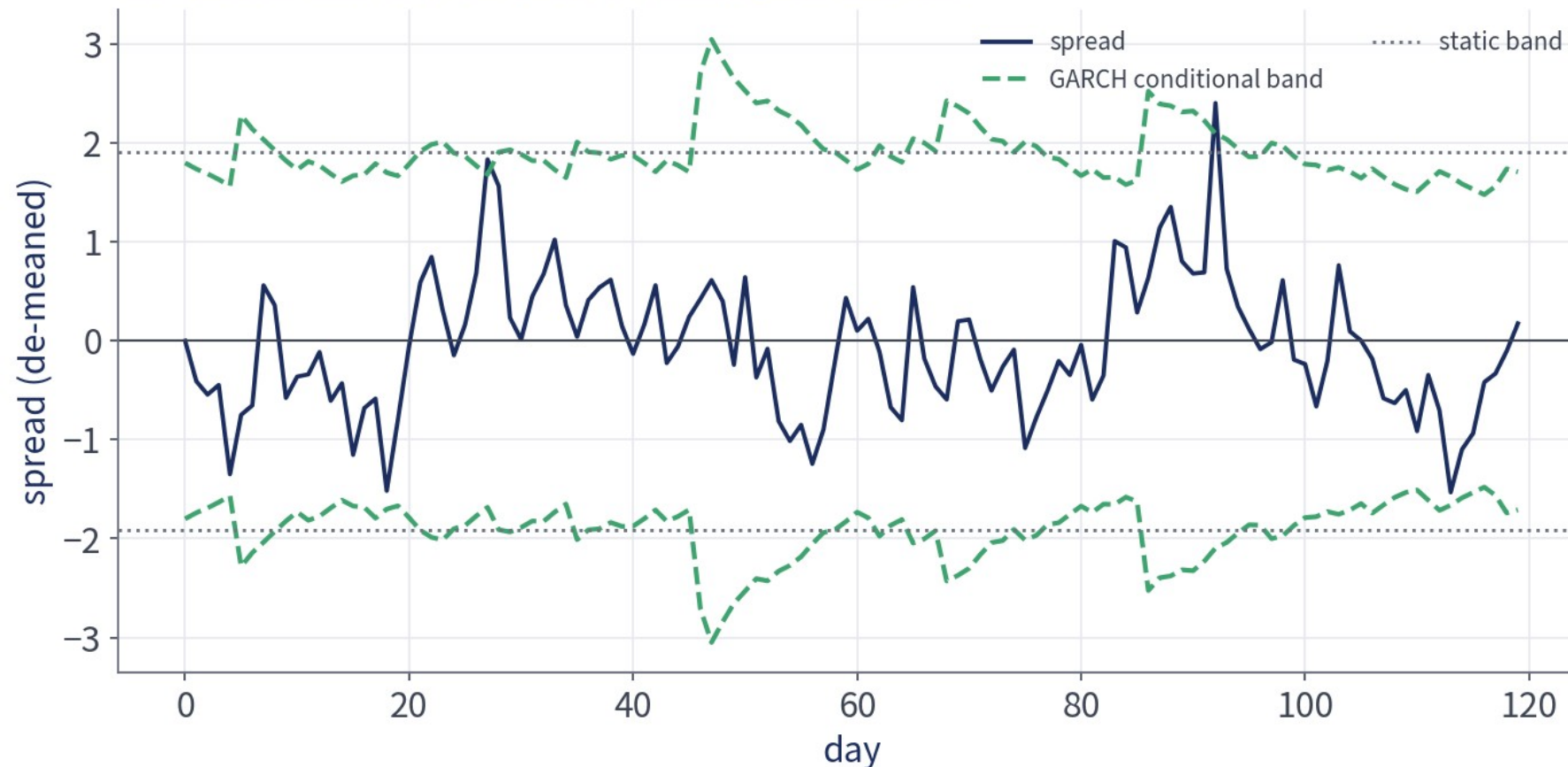
The payoff

- Wider bands when the spread is turbulent.
- Tighter bands when it's calm.
- Fewer false signals, better-timed trades.

Bands That Move With Risk

The GARCH conditional bands (green) breathe with the spread's volatility - flaring out during turbulent stretches and drawing in when it's quiet. Static bands (dotted) stay fixed, triggering too often in calm and too rarely in storms. Scaling with risk times the pairs trade better.

Case B: bands widen when the spread is volatile, tighten when calm



Trust, but Verify

Validation and dynamic thresholds

Why back-test

- A VaR that breaches too often is false comfort.
- Kupiec turns the breach count into a verdict.
- Regulators and desks both demand it.

Why conditional bands

- The spread's risk isn't constant.
- Static bands misfire across regimes.
- GARCH bands adapt, trade by trade.

The Week's Arc

From uncovering a risk

Week 5 built a full pipeline. ARCH saw that variance is conditional; GARCH modeled it parsimoniously and made it forecastable; Value-at-Risk turned that forecast into a dollar figure; the Kupiec test validated it; and the same conditional volatility scaled a pairs trade's bands. One idea - risk that clusters and mean-reverts - runs

The risk number (Case A)

- 95% VaR \$247K, 99% \$349K.
- Expected Shortfall \$309K.
- It breathes with volatility.

The validation (Case B)

- Kupiec passes at 14 breaches, fails at 25.
- Conditional bands scale with the spread.
- Forecast, validated, and traded.

Reproduce the VaR and the Back-Test

Direct Claude Code to check your arithmetic

GOAL Compute VaR and ES from a GARCH forecast, then back-test with Kupiec.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - VaR and ES:  
  * with V=$10M and a 1.5% GARCH vol forecast,  
    print the 95% and 99% VaR;  
  * print the 95% Expected Shortfall.  
CASE B - back-test:  
  * for N=250 and breaches x=14 and x=25, print the  
    Kupiec LR statistic and compare to 3.84;  
  * build GARCH conditional bands on a spread.  
Then plot the dynamic VaR and the conditional bands.
```

Check against the slides

- 95% VaR \$247K, 99% \$349K.
- ES \$309K > VaR.
- Kupiec 0.18 pass, 10.33 reject.
- bands scale with vol.

Deliverable: part3_practice.py whose printed numbers match the VaR, the ES, and the Kupiec verdicts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does a GARCH forecast become VaR?

Volatility times a quantile times the position

Q2 Why is Expected Shortfall larger than VaR?

It averages the losses beyond the VaR line

Q3 What does the Kupiec test check?

Whether the VaR breach rate is statistically correct

All three come from the VaR formula, the Expected Shortfall, and the Kupiec back-test.

Three Pitfalls to Avoid

Where VaR and pairs trip people

Common mistakes

- "A static VaR is good enough."
- "VaR tells you the worst-case loss."
- "Fixed z-score bands suit every regime."

The reality

- Static VaR understates risk after shocks.
- VaR is a threshold; ES sizes the tail beyond it.
- Scale pairs bands with the spread's volatility.

Two Calculations to Remember

Part 3 applications in one screen

1

A forecast becomes a risk number

A 1.5% GARCH volatility forecast on a \$10M book gave a 95% VaR of \$247K and a 99% VaR of \$349K, with an Expected Shortfall of \$309K in the tail beyond. Because the volatility is conditional, the VaR breathes - rising after shocks, relaxing in calm.

2

Validate the number, then trade on it

The Kupiec test passed a model with 14 breaches (statistic 0.18) and rejected one with 25 (statistic 10.33). The same conditional volatility scaled a pairs trade's bands - wide when the spread is turbulent, tight when it's calm.

3

One idea runs through the week

Risk that clusters and mean-reverts drove ARCH, GARCH, Value-at-Risk, its back-test, and the pairs bands alike - a single volatility model turned into a working risk desk.

Week 5 complete — from ARCH and GARCH to a validated Value-at-Risk and a volatility-timed pairs trade.